

Results Report: Representation-Level Analysis of Rotation Invariance in Spiking and Analog $p4$ -Equivariant CNNs

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August 2026

Abstract

This report presents multi-seed (seeds 0, 1, 2) experimental results for four of the experiments proposed in the research plan “Spike Dynamics for Learning Invariant Representations”: layer-wise emergence of invariance (Experiment 5.1), rotation sensitivity of individual neurons across layers (Experiment 5.3), rotation sensitivity through time (Experiment 5.4), temporal evolution of invariance via accuracy at truncated T (Experiment 5.7), and direct equivariance error (Experiment 5.8). Results are reported for four spike-placement configurations sharing an identical $p4$ -equivariant backbone: ANALOG (no spiking layers), ALTERNATING (spikes at layers 2, 4, 6), FULL SPIKING (all 7 layers spike), and HYBRID LATE (only layer 6 spikes). All notation follows the research plan exactly, restated in Section 2 for self-containedness. The central empirical finding is that the layer-wise and temporal signatures of rotation sensitivity are shaped substantially by the $p4$ -equivariant architecture itself, independently of spike placement, while spike placement modulates *how* and *when* that sensitivity manifests, in ways that do not always follow the direction one would naively predict.

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1 How to Read This Report

This report has three parts. Section 2 restates, without alteration, the notation defined in the research plan “Spike Dynamics for Learning Invariant Representations” — every symbol used later is defined here first, in the same form KV sir’s document uses it, so this report can be read on its own or side by side with the research plan. Section 3 restates the experimental setup. Sections 5–8 present the actual multi-seed results, organized by which of the nine proposed experiments (Section 5 of the research plan) they answer. Figures could not be embedded in this pass; every figure has a labelled placeholder box giving its exact filename on the GPU server (`gpuserver.cmi.ac.in`), so the corresponding PNG can be inserted later without losing track of which plot goes where.

2 Notation for Internal Representations

This section restates Section 4 of the research plan exactly, together with the additional quantities (direct equivariance error, truncated- T accuracy) needed for Sections 7 and 8. Every symbol that appears in a later plot or table is defined here before it is used.

2.1 Layers, Inputs, and Rotations

The network consists of layers

$$x \rightarrow L_1 \rightarrow L_2 \rightarrow \cdots \rightarrow L_L, \quad L = 7,$$

where x is a 28×28 input digit image (class 3, 4, 8, or 9), shown upright during training. Given a rotation angle θ , $R_\theta x$ denotes the same image rotated by θ . Layer index ℓ runs from 1 (nearest the input) to $L = 7$ (nearest the classifier); all layer-wise plots in this report place ℓ on the horizontal axis running shallow $\rightarrow$ deep.

All four configurations examined here (Section 3) share the identical $p4$ -equivariant backbone; they differ only in which layers use spiking (LIF) neurons versus analog (ReLU) activations.

2.2 Layer Representations

For an input image x , $h_\ell(x)$ denotes the internal representation after layer ℓ . In a conventional (analog) neural network, $h_\ell(x)$ is simply the vector of activations of all neurons in layer ℓ .

In a spiking neural network each neuron evolves over time, so there are several possible notions of representation. Throughout this work, following the research plan, we use the *accumulated spike counts* over the simulation horizon T . Suppose layer ℓ contains n_ℓ neurons and neuron i produces the spike train

$$S_i(1), S_i(2), \dots, S_i(T).$$

Its accumulated spike count is

$$a_i(x) = \sum_{t=1}^T S_i(t). \tag{1}$$

The layer representation is therefore

$$h_\ell(x) = (a_1(x), a_2(x), \dots, a_{n_\ell}(x)),$$

the vector of spike counts of all neurons in layer ℓ . For analog layers, $h_\ell(x)$ is the ordinary activation vector; these are treated as a degenerate spiking layer with $T = 1$ so that the same notation applies uniformly across configurations.

Why this matters. The central question of the whole project is whether $h_\ell(x)$ becomes increasingly invariant to rotation ($h_\ell(x) \approx h_\ell(R_\theta x)$) as ℓ increases. Everything in Section 5 is built directly on this comparison.

2.3 Neuron Responses

For an individual neuron i , $a_i(x)$ denotes its response to input x . Unlike a conventional neuron, an SNN neuron produces an entire time series consisting of membrane potentials $U_i(1), \dots, U_i(T)$ and spike outputs $S_i(1), \dots, S_i(T)$. Several choices are possible for defining the neuron’s scalar activation (accumulated spike count, average firing rate, final membrane potential, average membrane

potential); since classification throughout this work is based on accumulated spike counts, we use $a_i(x) = \sum_{t=1}^T S_i(t)$ (Eq. 1) uniformly, as specified in the research plan.

2.4 Representation Similarity

To study where invariant representations emerge, we compare the layer representations of an image and its rotated version, $h_\ell(x)$ versus $h_\ell(R_\theta x)$, using three similarity measures. High similarity indicates that layer ℓ has learned a representation that is insensitive to rotation.

Cosine similarity. Treats $h_\ell(x)$ and $h_\ell(R_\theta x)$ as vectors and measures the cosine of the angle between them (1 = identical direction, 0 = unrelated, -1 = opposite). This is the strictest of the three measures: it requires exact neuron-by-neuron agreement in both direction and relative magnitude, with no tolerance for the network re-using its neurons differently (e.g. permuting which unit encodes which feature) while preserving the same information content.

Centered Kernel Alignment (CKA). Compares the Gram matrix of pairwise similarities between a batch of images’ representations under $h_\ell(x)$ against the Gram matrix under $h_\ell(R_\theta x)$. Because CKA is invariant to orthogonal transformations and isotropic scaling of the neuron basis, it tolerates the network computing the same information in a different internal coordinate system, and is standard in the representation-similarity literature for this reason. *Implementation note:* for layer 1, the naive neuron-space Gram matrix has dimension $\sim 27,040 \times 27,040$ and is infeasible to materialize; we use the mathematically equivalent batch-side Gram matrix formulation instead, verified to agree exactly with the naive form on small test cases.

Singular Vector Canonical Correlation Analysis (SVCCA). Projects each representation onto its top principal directions, then finds the best linear alignment between the two reduced spaces and reports the resulting correlation. SVCCA tolerates *any* invertible linear transform relating the two representations, making it the most permissive of the three measures. *Caution:* SVCCA is known to report spuriously high similarity when the batch size is too small relative to the neuron count. A minimum batch-to-neuron ratio of $10\times$ is enforced throughout; even so, independent (genuinely dissimilar) data was empirically found to produce a baseline SVCCA score of roughly 0.2–0.4, not 0. SVCCA values should therefore be read *relatively* — larger is more similar — never as an absolute 0-to-1 invariance scale.

Metric	Tolerates	Strict about	Absolute scale?
Cosine similarity	Nothing — exact match required	Neuron identity, magnitude, direction	Yes
CKA	Rotation/rescaling of neuron basis	Pairwise similarity structure	Mostly yes
SVCCA	Any invertible linear transform	Top principal (signal) directions	No — compare relatively

Table 1: The three representation-similarity measures used in Experiment 5.1.

(Section 6) identifies where invariant representations are formed, at the resolution of individual neurons rather than whole layers.

2.5 Rotation Sensitivity of Individual Neurons

Following the research plan exactly, for neuron i we define the rotation sensitivity

$$\mathcal{R}_i = \text{Var}_\theta(a_i(R_\theta x)), \quad (2)$$

where the variance is computed over the collection of 24 rotation angles θ . A neuron with $\mathcal{R}_i \approx 0$ responds similarly to all rotations and is therefore approximately rotation-invariant; a large $\mathcal{R}_i$ indicates the neuron is strongly orientation-selective. Studying the distribution of $\mathcal{R}_i$ across layers (Section 6) identifies where invariant representations are formed, at the resolution of individual neurons rather than whole layers.

2.6 Temporal Representations and Rotation Sensitivity Through Time

Instead of collapsing spike trains into accumulated spike counts, we may consider the instantaneous representation

$$h_\ell(t, x) = (S_1(t), S_2(t), \dots, S_{n_\ell}(t)),$$

recording the spike outputs of every neuron in layer ℓ at timestep t specifically. Comparing $h_\ell(t, x)$ with $h_\ell(t, R_\theta x)$ for successive t investigates whether temporal integration progressively builds rotation-invariant representations. Correspondingly, $\mathcal{R}_i(t)$ denotes the rotation sensitivity of Eq. 2 computed using only the spike output at timestep t (or, in the aggregate plots of Section 6.2, the median over neurons of this quantity per layer, plotted against t).

2.7 Direct Equivariance Error

Experiment 5.8 of the research plan measures equivariance directly at the network output rather than through an internal-representation proxy:

$$\|f(Rx) - \rho(R)f(x)\|.$$

For the classification setting used throughout this work, f is the network’s output class-probability vector and the group action on the output, $\rho(R)$, is trivial (rotating the input should not permute the class labels), so this reduces to the relative error

$$\text{Equivariance error}(\theta) = \frac{\|f(R_\theta x) - f(x)\|}{\|f(x)\|}. \quad (3)$$

A perfectly rotation-invariant classifier has equivariance error 0 at every θ . The $p4$ -equivariant architecture used here *guarantees* error exactly 0 at the four group angles $\theta \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ by construction (Section 3.1), and makes no guarantee at any other angle, since those rotations are not elements of the $p4$ group the convolution kernels are built from. Experiment 5.8 tests precisely whether error stays low everywhere (full continuous invariance — not claimed by this architecture) or dips to 0 only at the four group angles while rising in between (the theoretically expected pattern).

2.8 Truncated- T Accuracy

Experiment 5.7 of the research plan asks whether invariance emerges progressively as spikes accumulate, by evaluating representations after $T = 1, 5, 10, 20, 50, 100$ timesteps. In this report we use $T' \in \{1, 5, 10, 20, 50\}$: a single model is trained once using the full $T = 50$ timesteps, and at evaluation time its own recorded spike train is truncated to the first T' timesteps to decode a classification decision, without retraining. This measures how quickly the network’s rotation-robust accuracy structure becomes available as a function of accumulated spiking time.

3 Experimental Setup

Every result below is computed on the 4-class rotated-MNIST task (digits 3, 4, 8, 9), trained on upright ($\theta = 0^\circ$) images only and evaluated at $\theta \in \{0^\circ, 15^\circ, \dots, 345^\circ\}$ (24 angles), averaged or shown individually over three independent training seeds $\{0, 1, 2\}$, for each of four spike-placement configurations described in Section 3.

3.1 Architecture: $p4$ -Equivariant Backbone

All four configurations share an identical 7-layer $p4$ -equivariant CNN backbone (six valid 3×3 $p4$ -convolutions and one valid 4×4 $p4$ -convolution, 10 channels, one max-pool after layer 2, a group-pooling step collapsing the 4 orientation channels via max, and a final linear classifier), matching Cohen & Welling’s rotated-MNIST P4CNN specification. The group $p4$ consists of translations composed with 90° rotations; the convolution kernels are constructed so that rotating the input by a multiple of 90° produces an output that is provably (not approximately) rotated/permuted correspondingly. This gives exact built-in invariance at $\theta \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ by mathematical construction and no guarantee elsewhere — which is precisely the reason the evaluation sweep uses all 24 angles rather than only the four group angles: the interesting question is what happens *between* the guaranteed points.

Observation 1 (BatchNorm breaks equivariance). *Standard BatchNorm assigns each of the 4 orientation channels independent running statistics, which breaks the architecture’s equivariance guarantee once trained upright-only (verified: an untrained model is equivariant to $\sim 10^{-6}$; after training with standard BatchNorm, equivariance error rises to 0.1–0.2). A custom `P4BatchNorm2d`, sharing statistics and affine parameters across the 4 orientation channels, restores equivariance to $\sim 10^{-6}$ even after training. All results in this report use `P4BatchNorm2d` throughout.*

3.2 Configurations

All four configurations have identical architecture and parameter count (24,744 parameters); they differ only in which layers use spiking LIF neurons (via `snntorch`, fast-sigmoid surrogate gradient) versus analog ReLU activations.

Configuration	Spiking layers	Description
ANALOG	none	Zero-spiking control / ceiling reference
ALTERNATING	2, 4, 6	Spiking interleaved with analog throughout depth
FULL SPIKING	all 1–7	Maximal spiking
HYBRID LATE	6 only	Spiking confined to near the output

Table 2: The four spike-placement configurations analyzed in this report. HYBRID EARLY (layers 1–2) and HYBRID MID (layers 3–5), both implemented, have not yet been trained under the corrected setup and are not covered here.

3.3 Training and Evaluation

- **Input encoding.** Poisson rate coding over $T = 50$ timesteps (direct/constant-current encoding produced a degenerate fixed-rate oscillation at low T and was abandoned).

- **Task.** 4-class rotated-MNIST, digits $\{3, 4, 8, 9\}$ — chosen because none is rotationally self-symmetric under $90^\circ/180^\circ$.
- **Training.** Upright (0°) images only, Adam, learning rate 10^{-3} , batch size 64, 20 epochs, best checkpoint selected by 0° test accuracy.
- **Evaluation.** Full 24-angle sweep, $\theta \in \{0^\circ, 15^\circ, \dots, 345^\circ\}$, held-out test split.
- **Seeds.** Three independent training runs per configuration, seeds $\{0, 1, 2\}$.
- **Hardware.** `gpuserver.cmi.ac.in`, $2\times$ RTX 2080 Ti.

3.4 Relation to the Original Internship Report

The original report’s compute-constrained single-seed runs ($T = 30$, 5 epochs, CPU only) reported two dramatic mechanistic findings: FULL SPIKING flat at 51–57% accuracy across all angles, and HYBRID LATE collapsing to exact chance (25%) at non-group angles. Under the corrected setup above, neither replicates: FULL SPIKING reaches $\sim 99\%$ upright and a mean of $\sim 94\%$ across all angles; HYBRID LATE’s worst-case accuracy across all seeds and angles is 80.9%. Three candidate explanations (the `P4BatchNorm2d` fix, the longer schedule, the higher T) have been identified but not yet isolated by controlled ablation (Section 10). The representation-level experiments in Sections 5–8 are a separate, complementary investigation of the same underlying question posed by the research plan (Section 1): even where accuracy differences between configurations have mostly disappeared, does the *internal representation* still reveal a meaningful role for spike dynamics?

4 Baseline: Multi-Seed Accuracy Comparison

Before the representation-level results, we restate the accuracy picture with all three seeds and error bars, since this is the backdrop against which Sections 5–8 should be read.

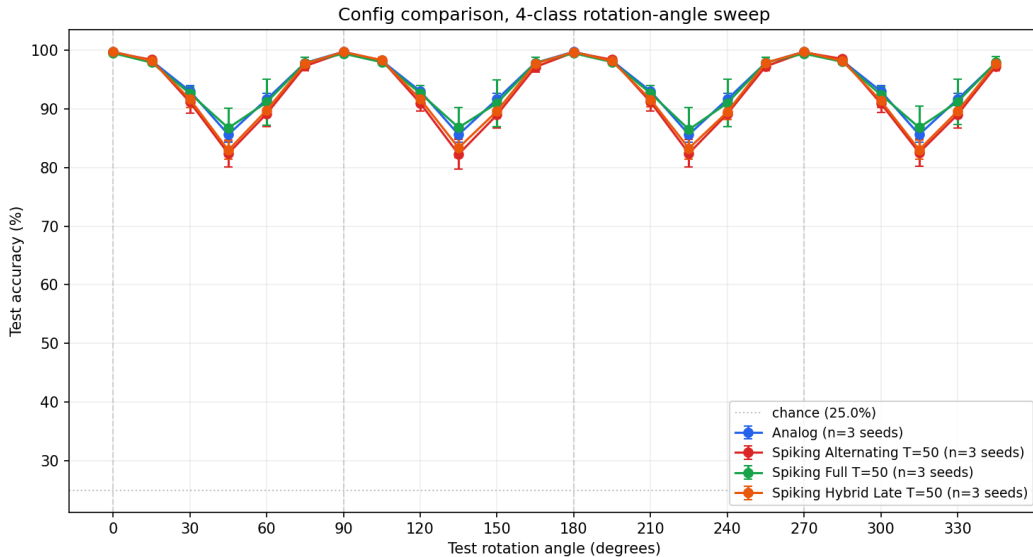


Figure 1: Config comparison, 4-class rotation-angle sweep. Mean $\pm$ std over $n = 3$ seeds for each of ANALOG, ALTERNATING, FULL SPIKING, HYBRID LATE. Chance level (25%) shown for reference. `n4_comparison_final.png`

With all three seeds and error bars, the four configurations’ rotation-accuracy curves sit within a few percentage points of one another at every angle, with substantial error-bar overlap at the hardest angles ($45^\circ, 135^\circ, 225^\circ, 315^\circ$). This confirms that the original report’s two dramatic accuracy-level findings (Section 3) do not replicate under the corrected setup, and that with proper multi-seed averaging the remaining accuracy differences between configurations are small relative to seed-to-seed variation.

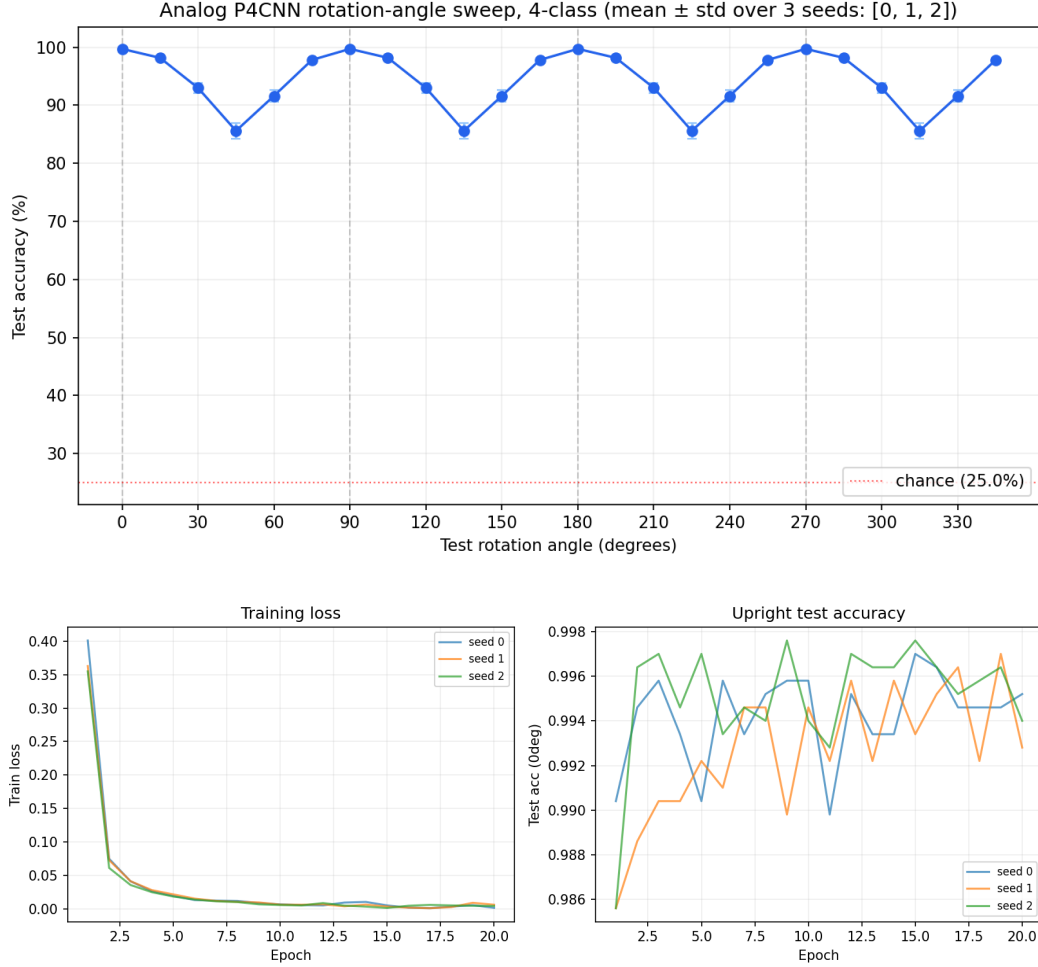


Figure 2: ANALOG-only rotation curve, mean $\pm$ std over 3 seeds ($\{0,1,2\}$), alongside per-seed training-loss and upright-accuracy curves. `n4_rotation_curve.png` and `n4_rotation_curve_training_curves.png`

5 Finding 1 — Layer-wise Emergence of Invariance (Experiment 5.1)

5.1 Setup

Following Section 5.1 of the research plan exactly: for an image x and its rotated version $R_\theta x$, we compute $h_\ell(x)$ and $h_\ell(R_\theta x)$ at every hidden layer $\ell = 1, \dots, 7$, and measure similarity using cosine similarity, CKA, and SVCCA (Section 2.4), plotted against layer depth for each of the 24

test angles.

5.2 Result: A Common Depth-Wise Collapse in Every Configuration

Every configuration — including ANALOG, the zero-spiking control — shows the same qualitative shape: CKA similarity to the upright representation stays at or above roughly 0.97–0.99 through layers 1–5, and only then drops sharply in layers 6–7, to roughly 0.70–0.85 at the hardest angles (typically the non-group angles nearest a group angle: 15° , 30° , 45° , 60° , etc.). The last two layers are where representational invariance to rotation is lost; the first five preserve it almost perfectly, in every configuration tested.

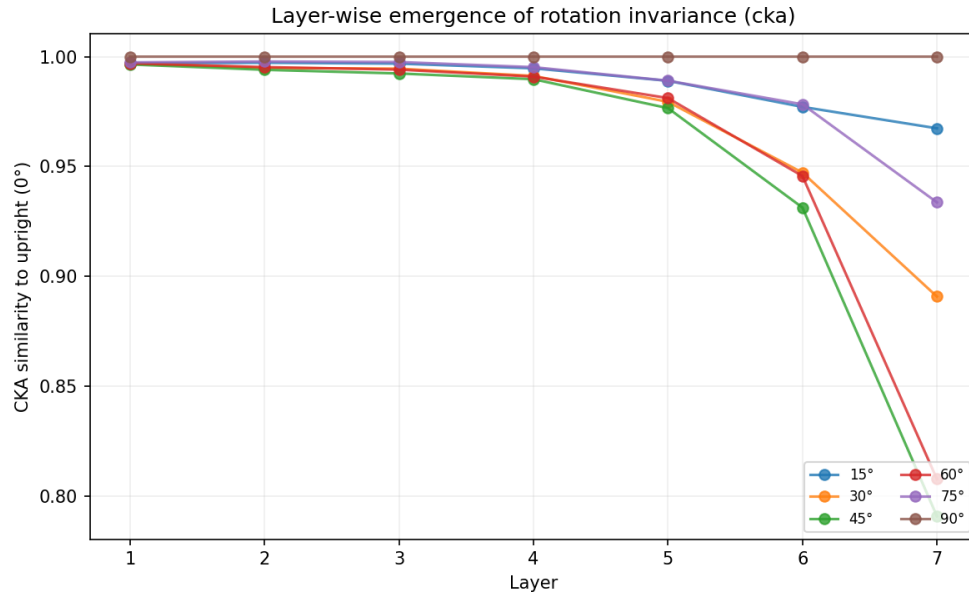


Figure 3: ANALOG — CKA similarity to upright vs. layer, one line per test angle. `analog_layerwise_cka.png`

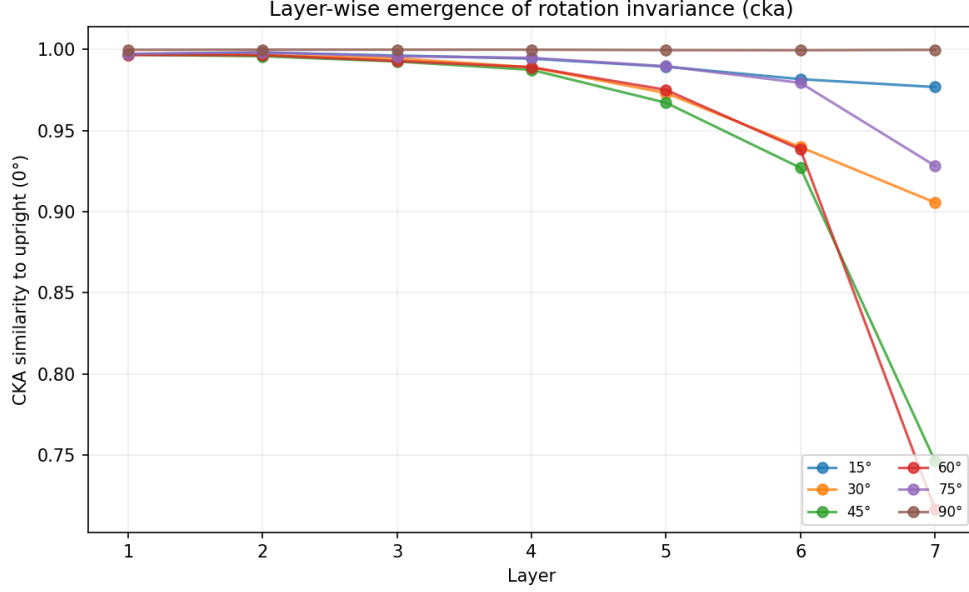


Figure 4: ALTERNATING — CKA similarity to upright vs. layer, one line per test angle. `alternating_layerwise_cka.png`

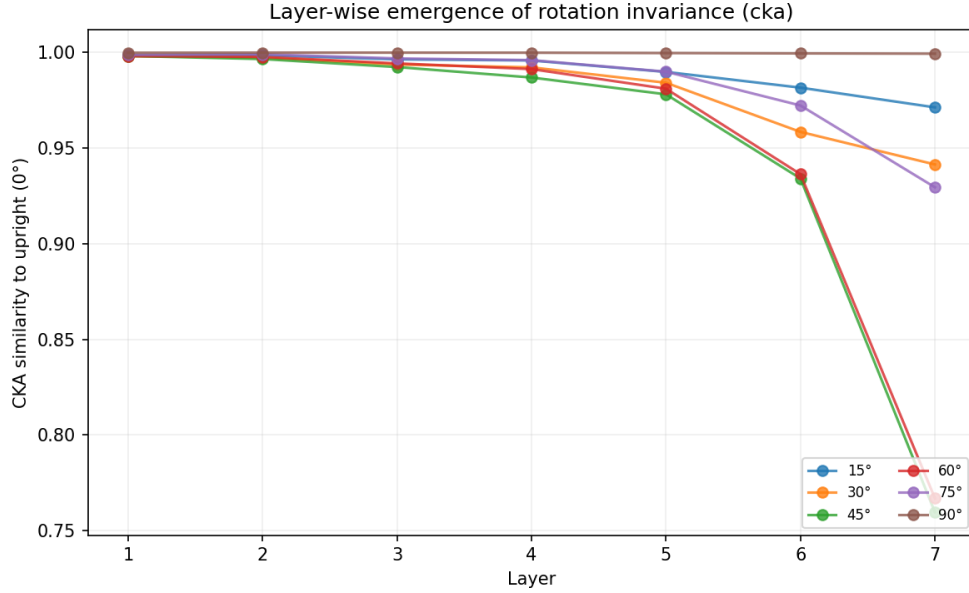


Figure 5: FULL SPIKING — CKA similarity to upright vs. layer, one line per test angle. `full_layerwise_cka.png`

Aggregate CKA summary (one line per angle, per configuration) HYBRID LATE — CKA similarity to upright vs. layer, one line per test angle. `hybrid_late_layerwise_cka.png` was not generated as a standalone aggregate for this pass; see the per-seed Section 5.1 plots below for HYBRID LATE instead.

5.3 Per-Seed, All-Three-Metrics Detail

The figures below show, for each configuration and seed, all three similarity measures side by side with all 24 angles overlaid, confirming the layer-6/7 collapse is not an artifact of one metric and is stable across seeds.

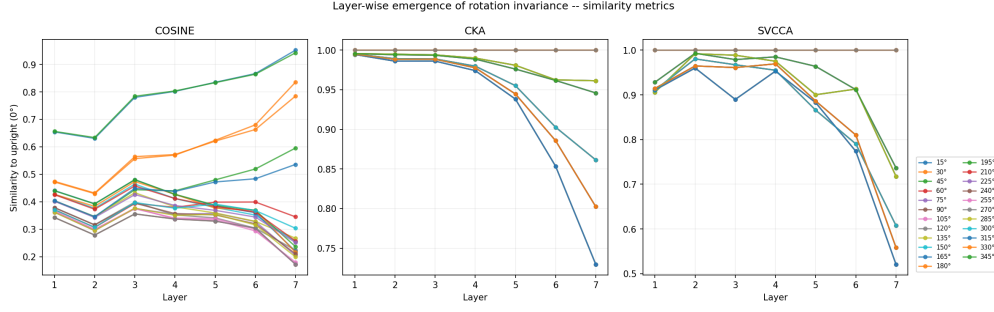


Figure 6: ANALOG, seed 0 — cosine / CKA / SVCCA vs. layer, all 24 angles. [analog_seed0/analog_seed0_5.1_layerwise_similarity.png](#)

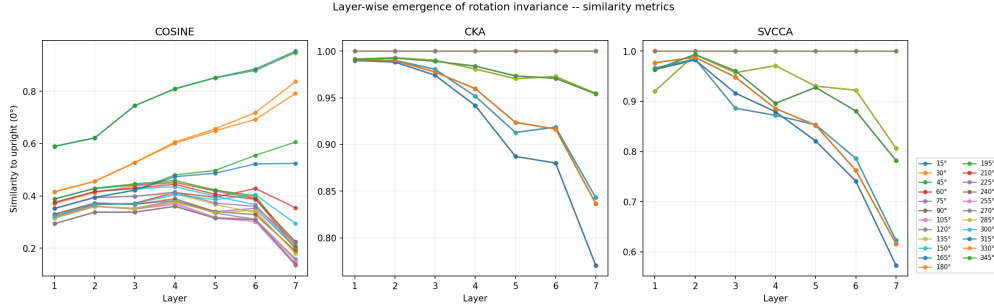


Figure 7: ANALOG, seed 1. [analog_seed1/analog_seed1_5.1_layerwise_similarity.png](#)

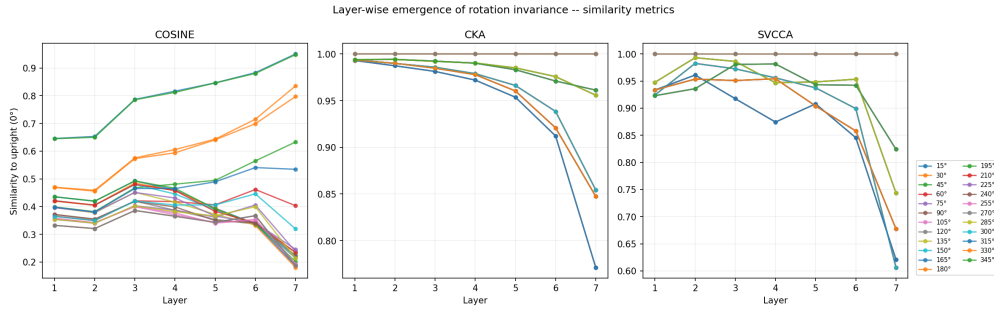


Figure 8: ANALOG, seed 2. [analog_seed2/analog_seed2_5.1_layerwise_similarity.png](#)

Analog

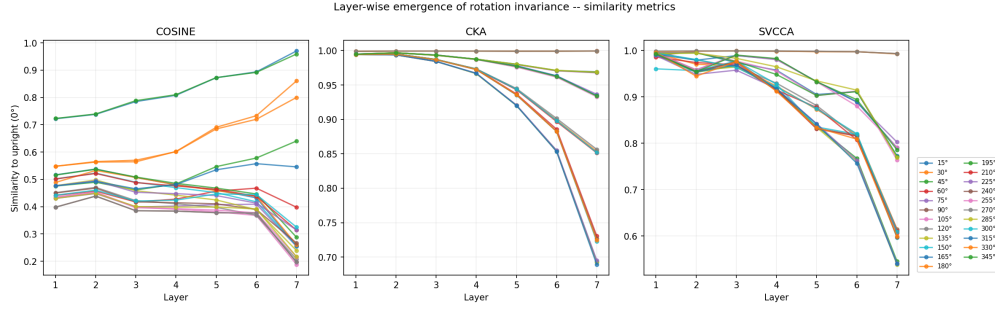


Figure 9: ALTERNATING, seed 0. [alternating_seed0/alternating_seed0_5.1_layerwise_similarity.png](#)

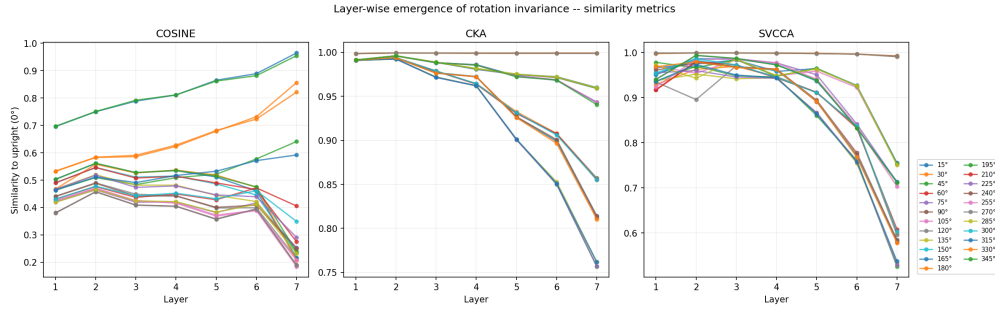


Figure 10: ALTERNATING, seed 1. [alternating_seed1/alternating_seed1_5.1_layerwise_similarity.png](#)

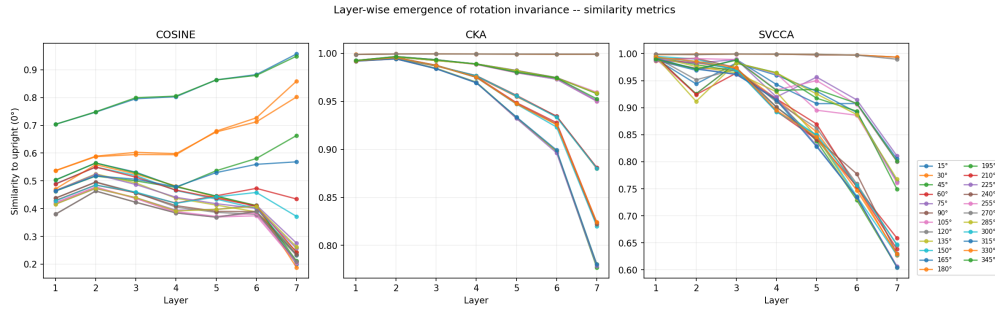


Figure 11: ALTERNATING, seed 2. [alternating_seed2/alternating_seed2_5.1_layerwise_similarity.png](#)

Alternating (spikes at layers 2, 4, 6)

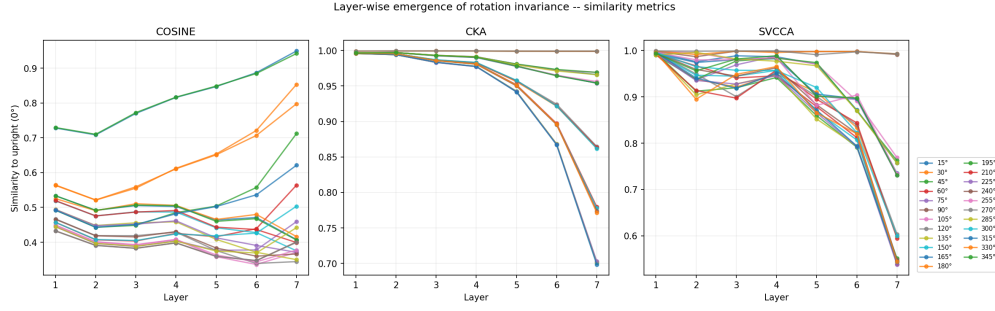


Figure 12: FULL SPIKING, seed 0. `full_seed0/full_seed0_5.1_layerwise_similarity.png`

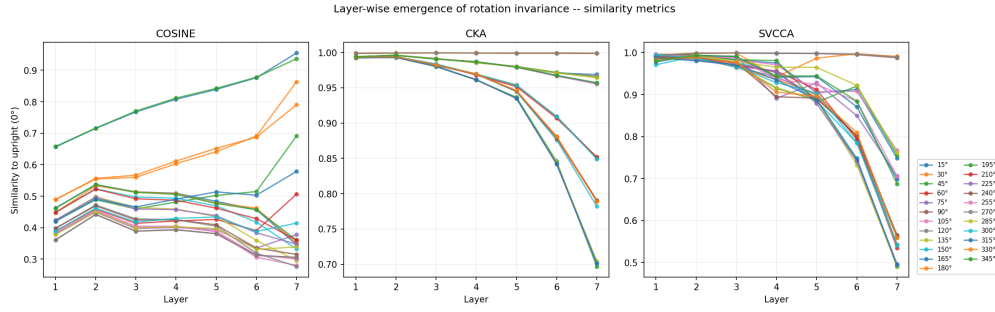


Figure 13: FULL SPIKING, seed 1. `full_seed1/full_seed1_5.1_layerwise_similarity.png`

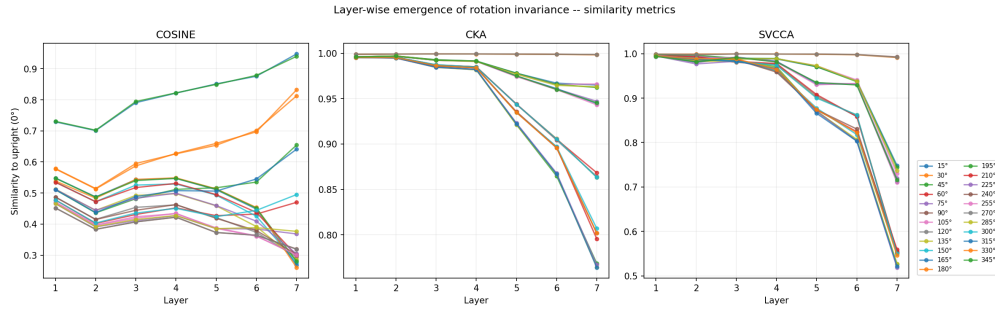


Figure 14: FULL SPIKING, seed 2. `full_seed2/full_seed2_5.1_layerwise_similarity.png`

Full Spiking (all 7 layers spike)

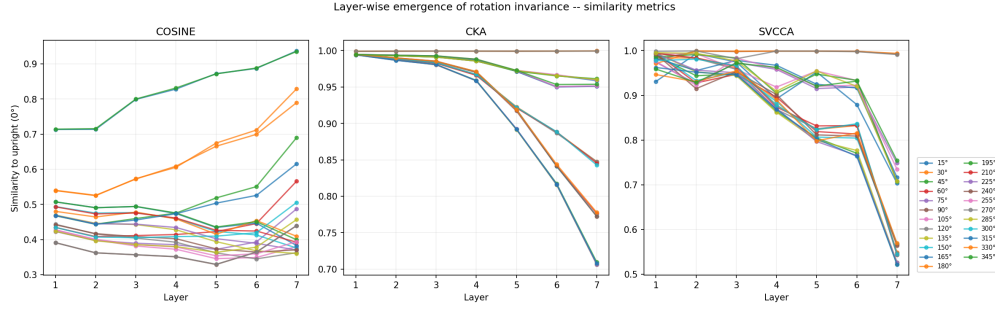


Figure 15: HYBRID LATE, seed 0. [hybrid_late_seed0/hybrid_late_seed0_5.1_layerwise_similarity.png](#)

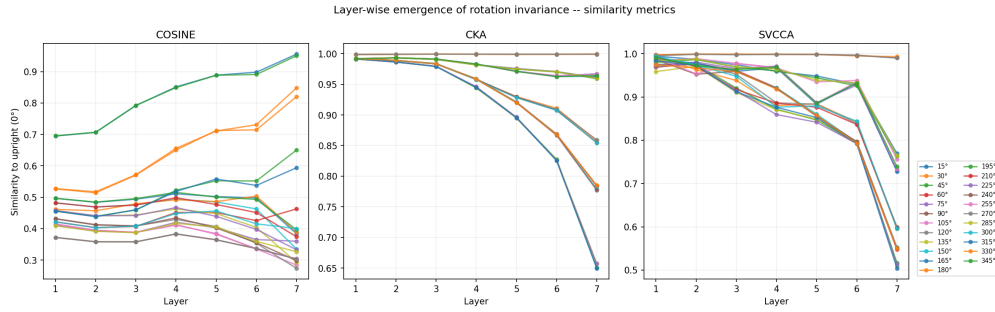


Figure 16: HYBRID LATE, seed 1. [hybrid_late_seed1/hybrid_late_seed1_5.1_layerwise_similarity.png](#)

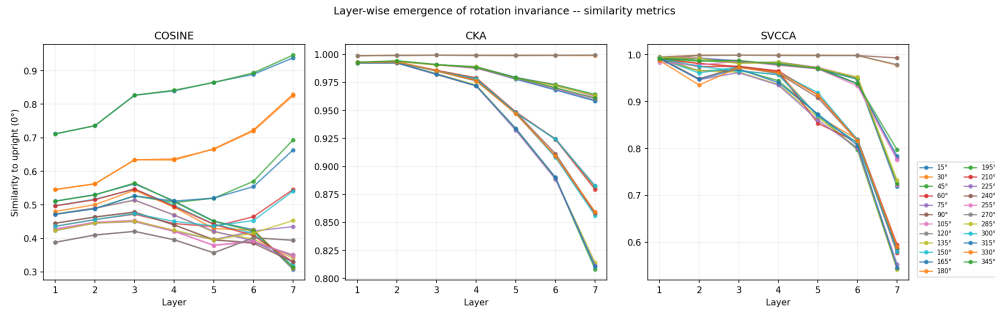


Figure 17: HYBRID LATE, seed 2. [hybrid_late_seed2/hybrid_late_seed2_5.1_layerwise_similarity.png](#)

Hybrid Late (layer 6 only spikes)

5.4 Reading the Three Metrics Together

The three panels within each figure disagree with each other informatively rather than redundantly. Cosine similarity is much lower and noisier throughout (typically 0.3–0.6 even at shallow layers, falling to 0.2–0.3 by layer 7), reflecting its strictness: it demands exact neuron-by-neuron agreement and penalizes even a meaningless internal reshuffling of which channel encodes which feature, even when the information content is unchanged. CKA and SVCCA both stay near 1.0 through layer 5 and only then diverge slightly at layers 6–7, with SVCCA sitting a little above CKA at the deepest layers in several seeds — consistent with SVCCA’s more permissive tolerance for any linear transform (Section 2.4) crediting structure that CKA’s stricter pairwise-similarity criterion does

not. The three metrics agreeing on *where* the drop happens while disagreeing on *how severe* it looks is itself informative: it indicates a real, robust change in what the layer computes, not an artifact of one particular similarity definition.

5.5 Interpretation

1. The representational collapse at layers 6–7 occurs in ANALOG too, with zero spiking neurons anywhere in the network. Whatever causes deep layers to become rotation-sensitive is a property of the $p4$ -equivariant architecture itself — most plausibly the group-pooling and final linear-classifier stages, the only components not built from strictly equivariant convolutions — not something introduced by spiking dynamics.
2. Spike placement modulates the depth pattern’s shape somewhat but not its existence or location. ALTERNATING and FULL SPIKING show a slightly earlier onset of divergence (visible already around layer 5) relative to ANALOG and HYBRID LATE, but all four configurations land in the same 0.70–0.85 CKA range by layer 7 at the hardest angles.
3. This complicates the research plan’s central hypothesis (Section 6 of the research plan, restated in Section 9 of this report) in an interesting way. The hypothesis space as originally framed treats spike dynamics as the primary lever for building (or failing to build) invariant representations. Finding 1 instead shows that the *architecture* sets both the ceiling and the location of where invariance is lost, and spike dynamics act as a secondary modulator on top of that. This motivates reframing the guiding question from “do spikes build invariance?” to “given that architecture alone determines where invariance breaks down, what does spike placement change about how that breakdown proceeds?” — precisely what Findings 2 and 3 investigate.

6 Finding 2 — Rotation Sensitivity of Individual Neurons, by Layer and by Time (Experiments 5.3, 5.4)

Finding 1 examined whole-layer representations. This section zooms into individual neurons using $\mathcal{R}_i$ (Eq. 2, Section 2.5), asking two complementary questions: which layers contain the most rotation-sensitive individual neurons (Experiment 5.3), and does $\mathcal{R}_i(t)$ decrease as more spiking timesteps are accumulated, as the research plan’s temporal-averaging hypothesis (Section 9.4 of the research plan) would predict (Experiment 5.4)?

6.1 5.3 — Distribution of $\mathcal{R}_i$ Across Layers

Each boxplot shows, for one configuration, the distribution of $\mathcal{R}_i$ across all neurons in each of the 7 layers (log scale, since $\mathcal{R}_i$ spans several orders of magnitude). The orange line is the median; the box spans the interquartile range; whiskers extend to the typical range with outliers beyond.

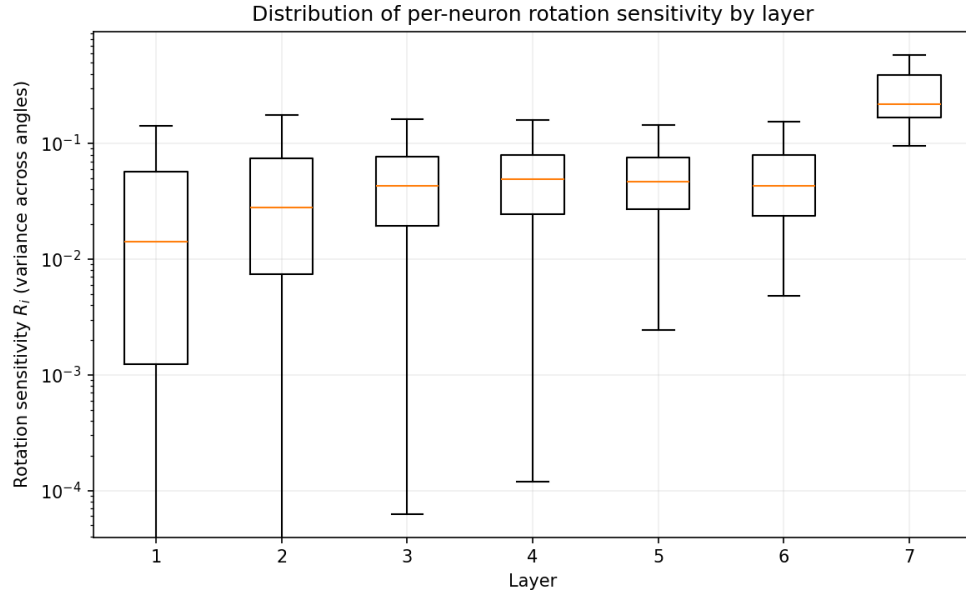


Figure 18: ANALOG — $\mathcal{R}_i$ distribution by layer. analog_rotation_sensitivity.png

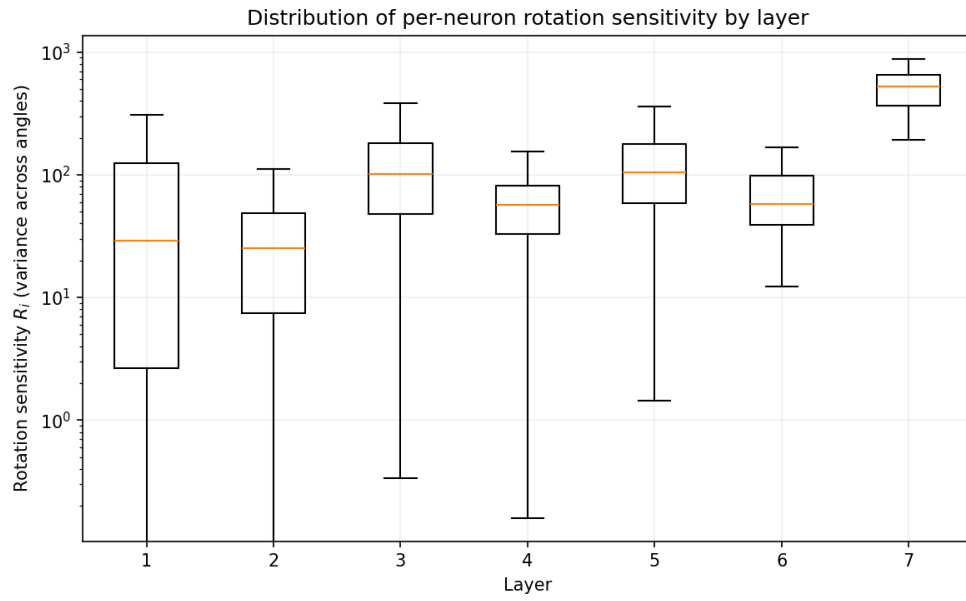


Figure 19: ALTERNATING — $\mathcal{R}_i$ distribution by layer. alternating_rotation_sensitivity.png

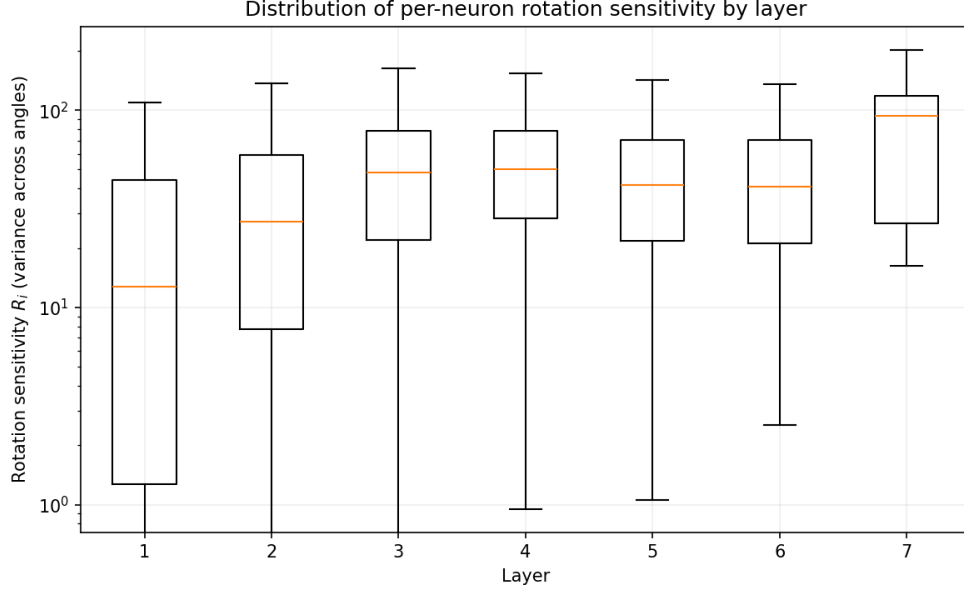


Figure 20: FULL SPIKING — $\mathcal{R}_i$ distribution by layer. `full_rotation_sensitivity.png`

Aggregate boxplots (seeds pooled per configuration) HYBRID LATE — $\mathcal{R}_i$ distribution by layer. `hybrid_late_rotation_sensitivity.png` was not generated as a standalone aggregate for this pass; see the per-seed plots below for HYBRID LATE instead.

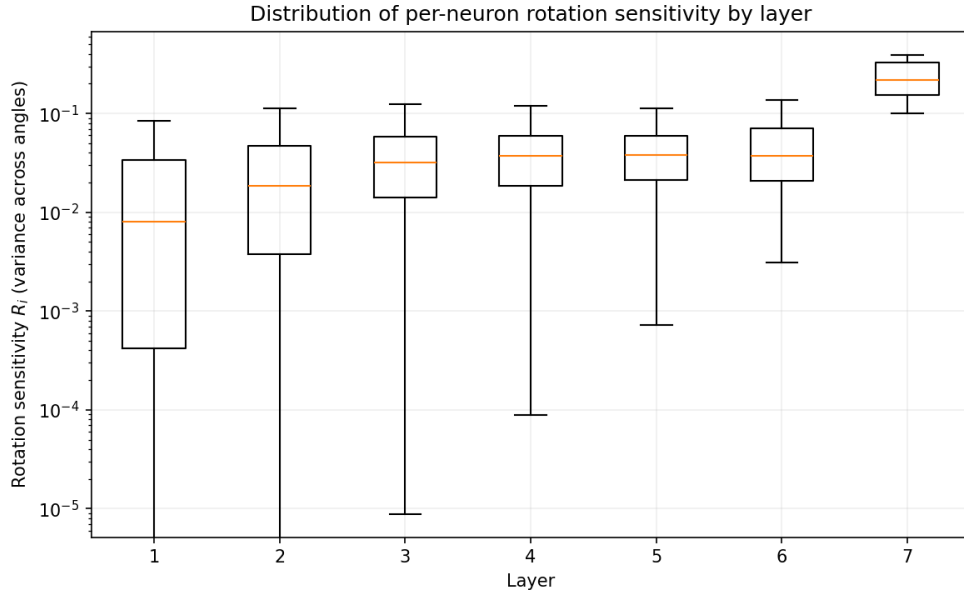


Figure 21: ANALOG, seed 0. `analog_seed0/analog_seed0_5.3_rotation_sensitivity.png`

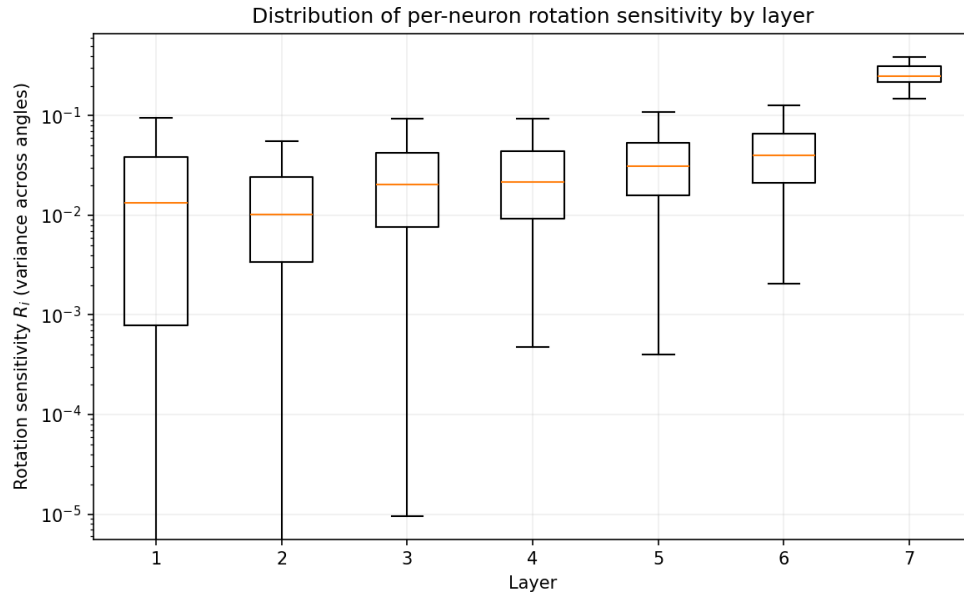


Figure 22: ANALOG, seed 1. analog_seed1/analog_seed1_5.3_rotation_sensitivity.png

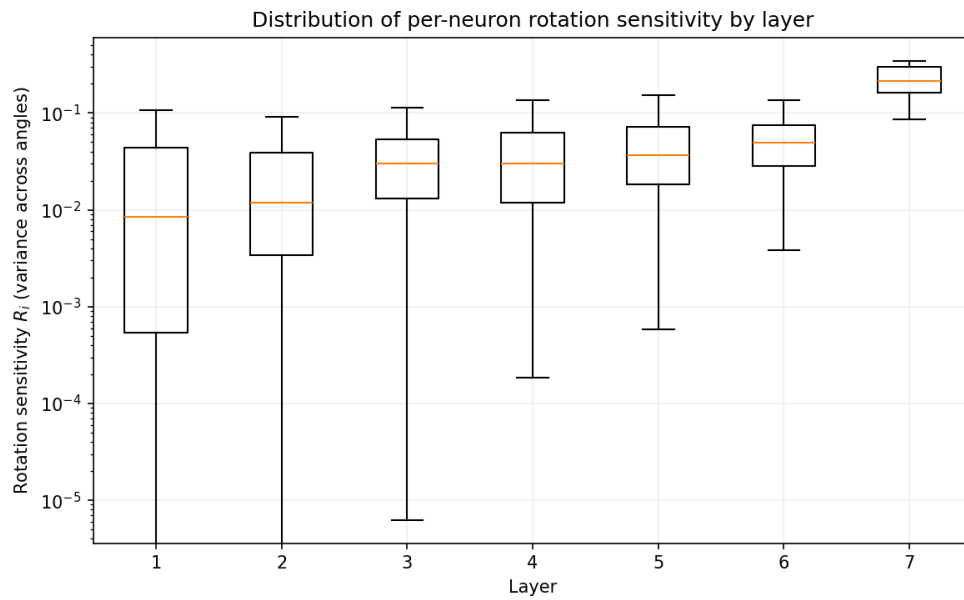


Figure 23: ANALOG, seed 2. analog_seed2/analog_seed2_5.3_rotation_sensitivity.png

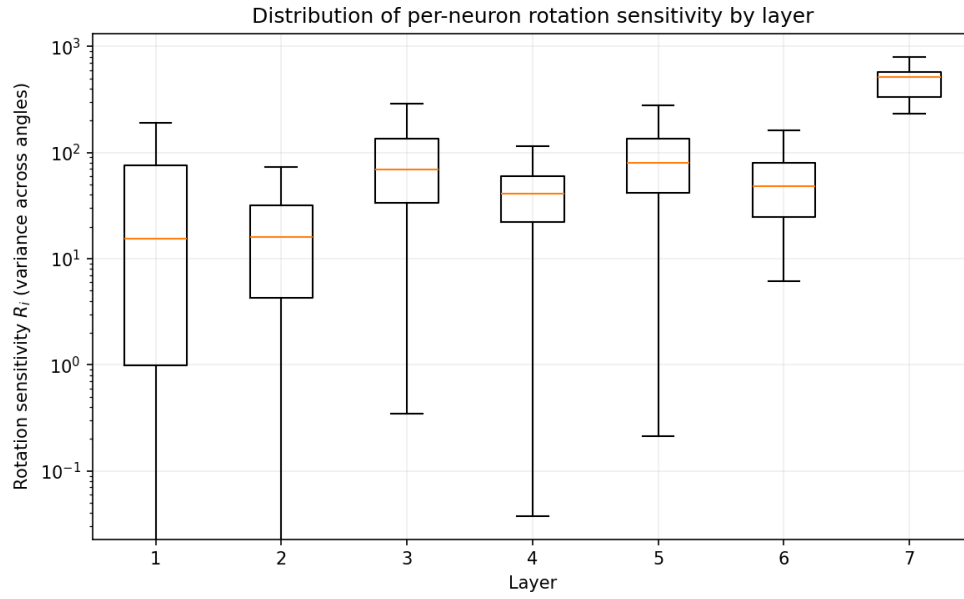


Figure 24: ALTERNATING, seed 0. alternating_seed0/alternating_seed0_5.3_rotation_sensitivity.png

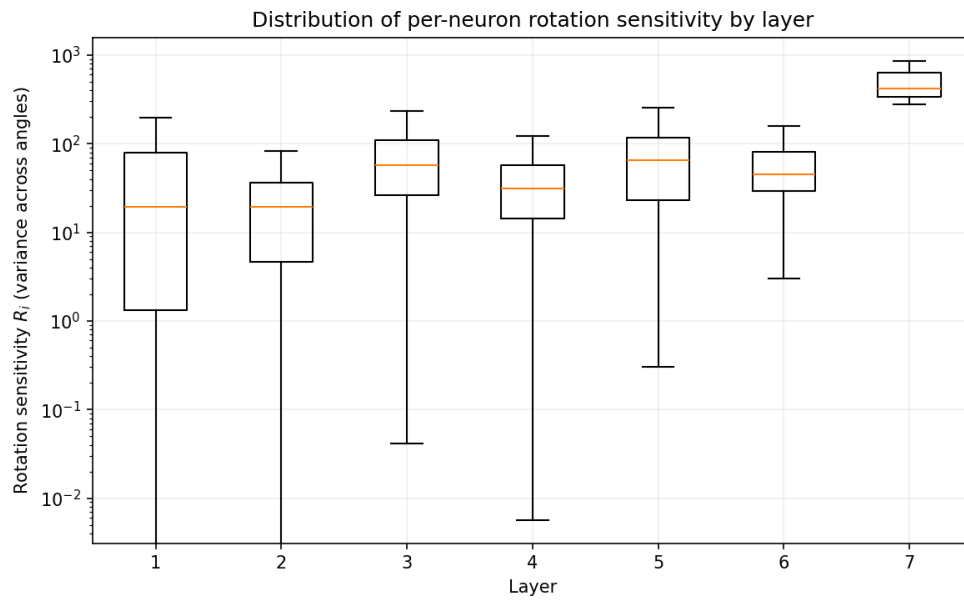


Figure 25: ALTERNATING, seed 1. alternating_seed1/alternating_seed1_5.3_rotation_sensitivity.png

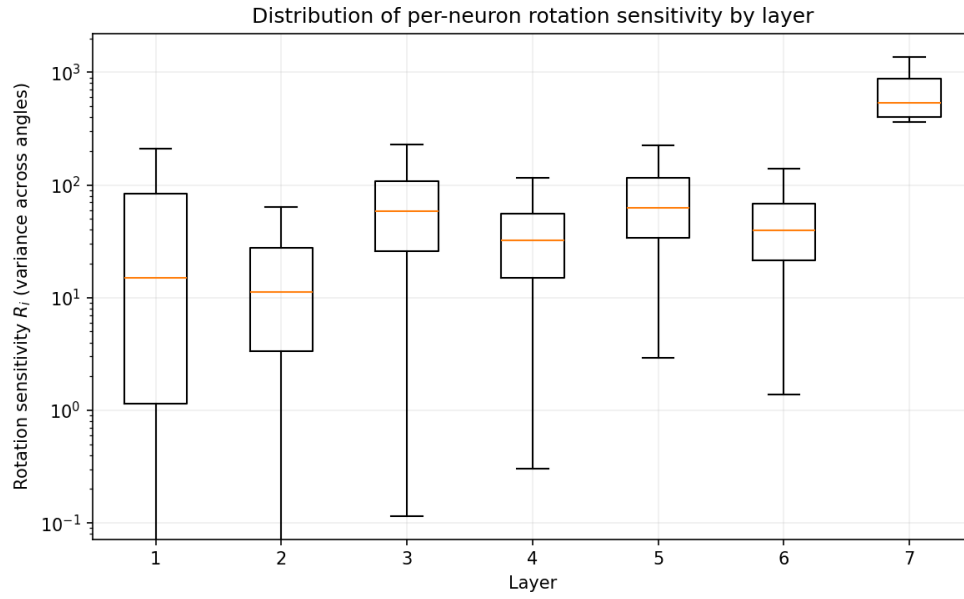


Figure 26: ALTERNATING, seed 2. `alternating_seed2/alternating_seed2_5.3_rotation_sensitivity.png`

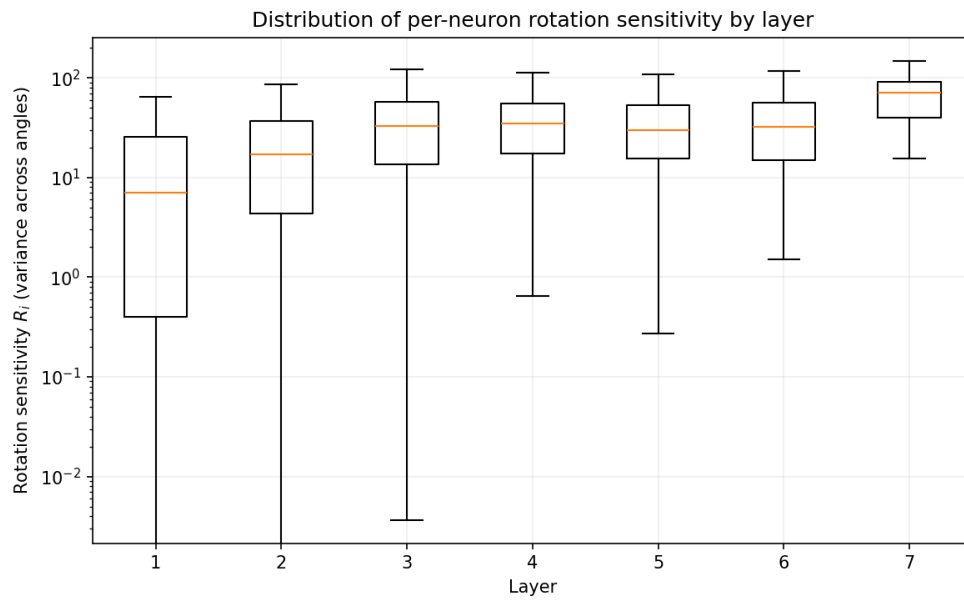


Figure 27: FULL SPIKING, seed 0. `full_seed0/full_seed0_5.3_rotation_sensitivity.png`

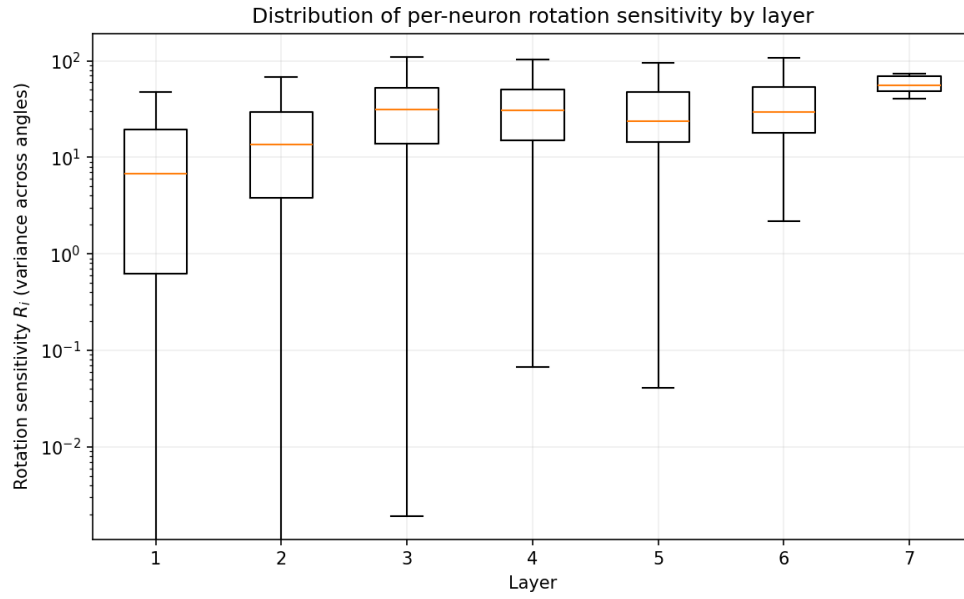


Figure 28: FULL SPIKING, seed 1. `full_seed1/full_seed1_5.3_rotation_sensitivity.png`

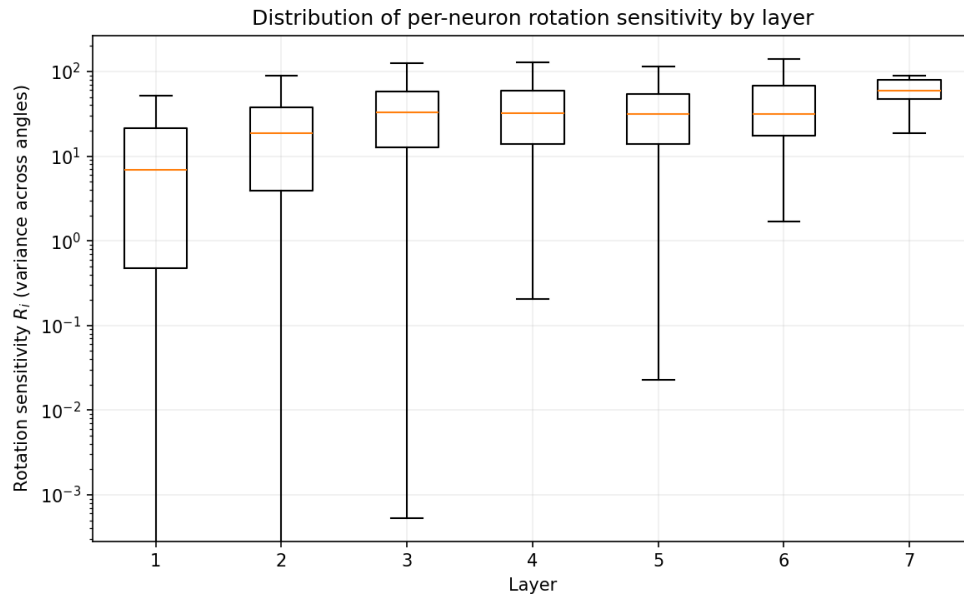


Figure 29: FULL SPIKING, seed 2. `full_seed2/full_seed2_5.3_rotation_sensitivity.png`

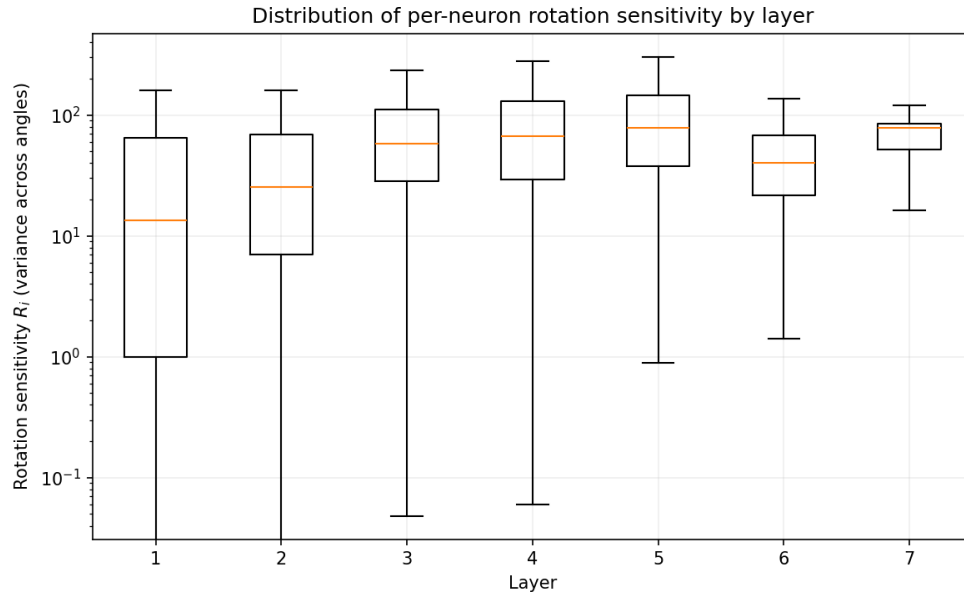


Figure 30: HYBRID LATE, seed 0. `hybrid_late_seed0/hybrid_late_seed0_5.3_rotation_sensitivity.png`

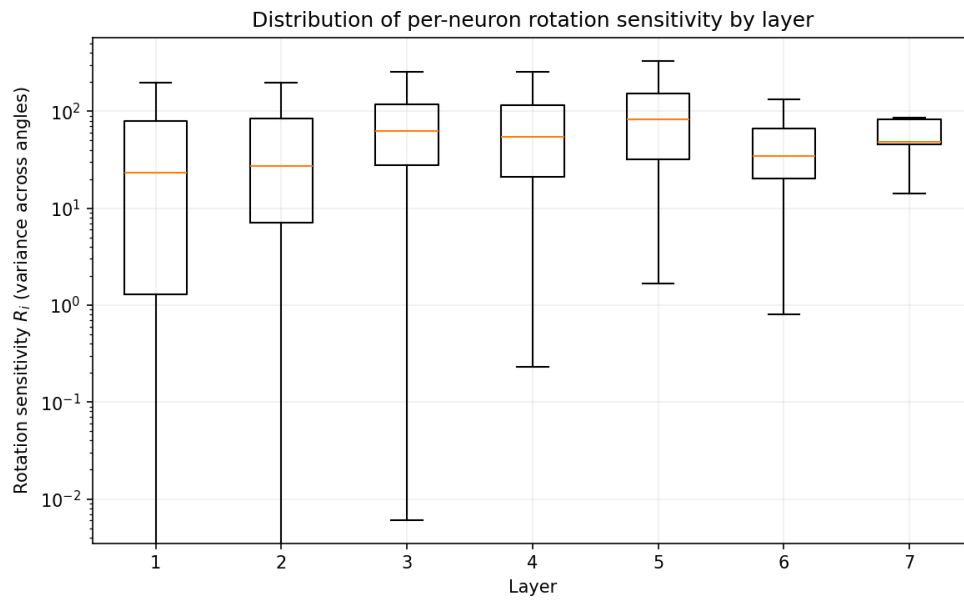


Figure 31: HYBRID LATE, seed 1. `hybrid_late_seed1/hybrid_late_seed1_5.3_rotation_sensitivity.png`

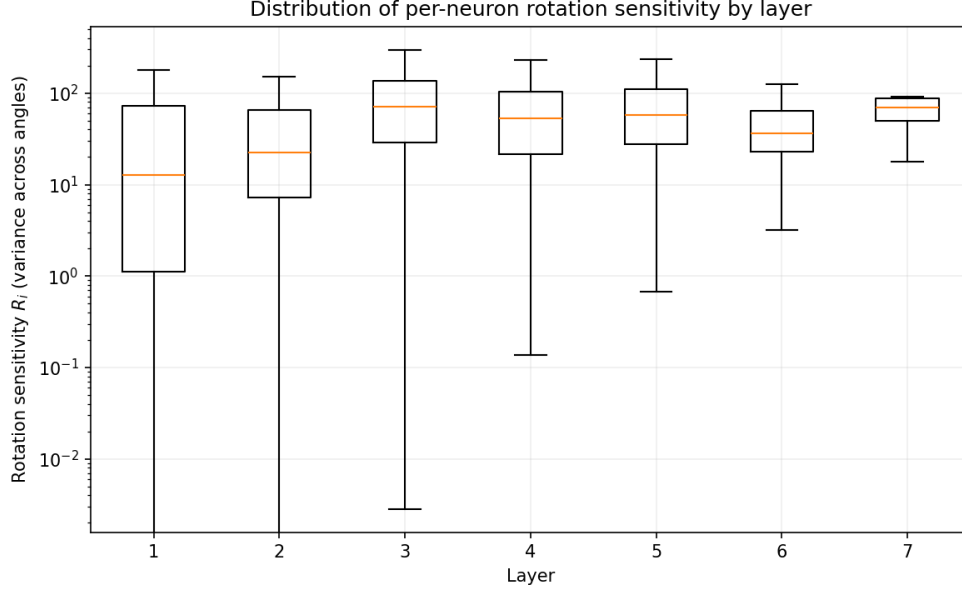


Figure 32: HYBRID LATE, seed 2. `hybrid_late_seed2/hybrid_late_seed2_5.3_rotation_sensitivity.png`

Per-seed boxplots

What differs between configurations All four configurations place layer 7 at or near the top of the $\mathcal{R}_i$ ranking, consistent with Finding 1’s identification of the last layers as where invariance breaks down. The shape of the climb to layer 7 differs meaningfully:

- **Alternating:** a sharp, order-of-magnitude jump only at layer 7 — median $\mathcal{R}_i \sim 10\text{--}100$ for layers 1–6, jumping to $\sim 400\text{--}800$ at layer 7.
- **Full Spiking:** a gradual, full-depth climb — median $\mathcal{R}_i$ increases roughly monotonically from ~ 10 (layer 1) to $\sim 90\text{--}100$ (layer 7), with no single sharp jump.
- **Hybrid Late:** nearly flat across all seven layers — median $\mathcal{R}_i$ sits in a comparatively narrow band ($\sim 10\text{--}80$) at every depth, including layer 7.
- **Analog:** a gradual full-depth climb similar in shape to FULL SPIKING, but at a dramatically smaller absolute scale ($\sim 0.01\text{--}0.05$ throughout, roughly $1000\times$ smaller than the spiking configurations at comparable layers).

Observation 2 ($\mathcal{R}_i$ scale is not comparable across activation types). *Because $\mathcal{R}_i$ ’s absolute scale depends on whether the underlying activation is a continuous ReLU output or a spike count, the fair cross-configuration comparison is the shape of the $\mathcal{R}_i$ -vs-layer curve (sharp late jump vs. gradual climb vs. flat), not the raw magnitude. Within a single configuration, comparisons across layers or across seeds are on safe footing.*

6.2 5.4 — Rotation Sensitivity Through Time

This experiment is the direct empirical test of the research plan’s Section 9.4 hypothesis: does temporal averaging progressively stabilize the representation, so that $\text{Var}(\bar{h}_T) = O(T^{-1})$ and $\mathcal{R}_i(t)$

decays toward 0 as t grows? The plots below show median $\mathcal{R}_i(t)$ per layer against timestep $t \in \{1, \dots, 50\}$, log scale on the vertical axis.

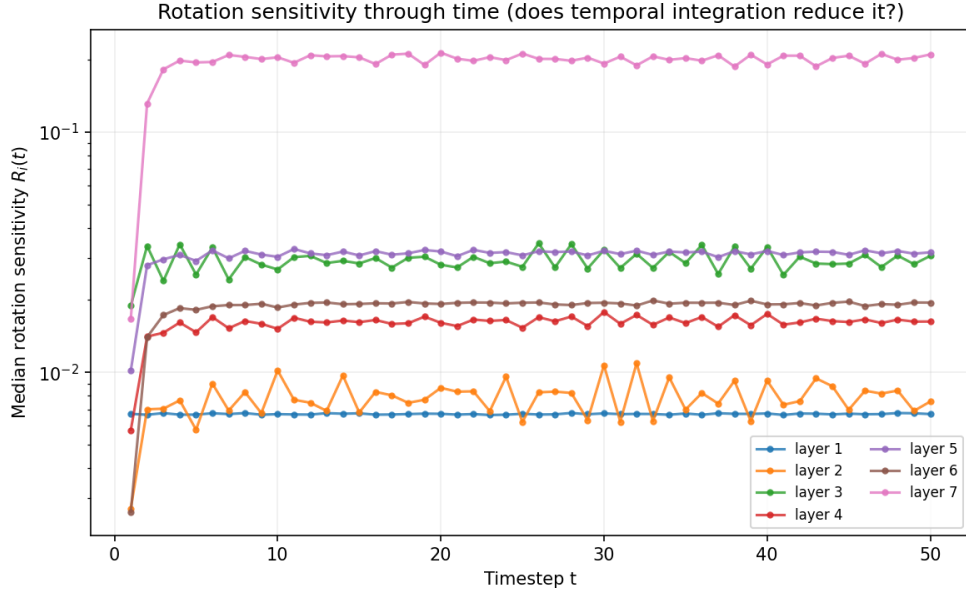


Figure 33: ALTERNATING, seed 0 — $\mathcal{R}_i(t)$ vs. timestep, all 7 layers. [alternating_seed0/alternating_seed0_5.4_rotation_sensitivity_through_time.png](#)

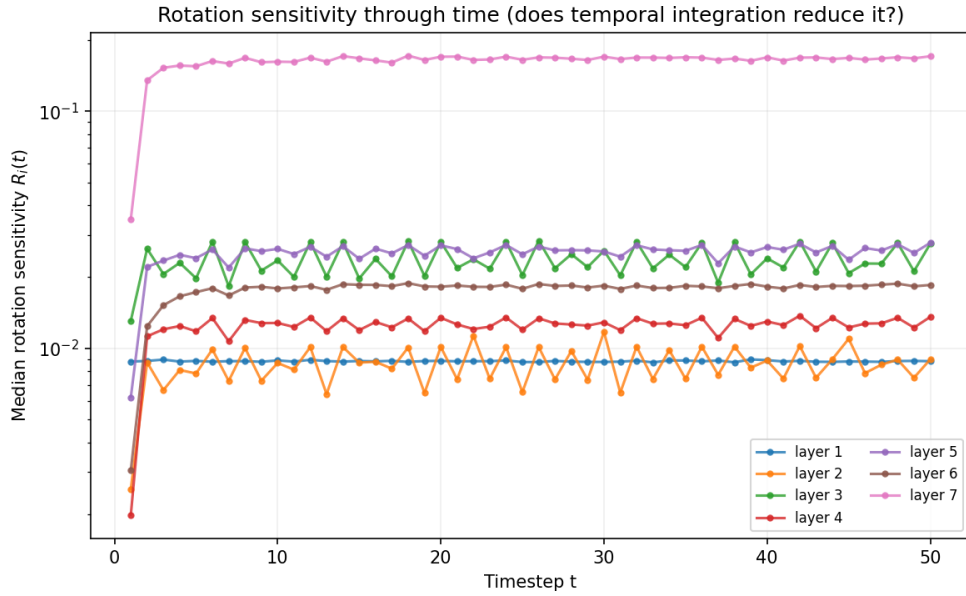


Figure 34: ALTERNATING, seed 1. [alternating_seed1/alternating_seed1_5.4_rotation_sensitivity_through_time.png](#)

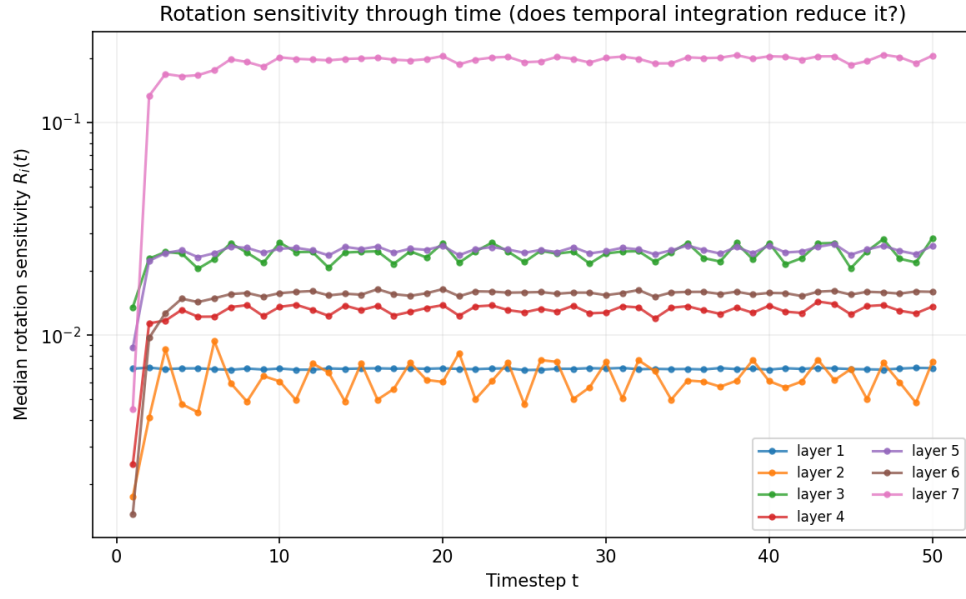


Figure 35: ALTERNATING, seed 2. alternating_seed2/alternating_seed2_5.4_rotation_sensitivity_through_time.png

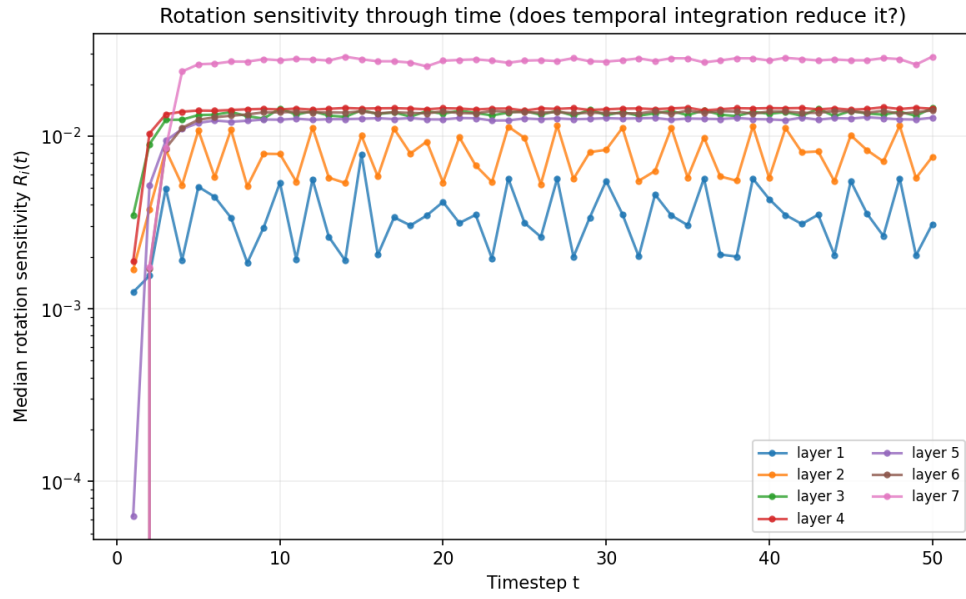


Figure 36: FULL SPIKING, seed 0. full_seed0/full_seed0_5.4_rotation_sensitivity_through_time.png

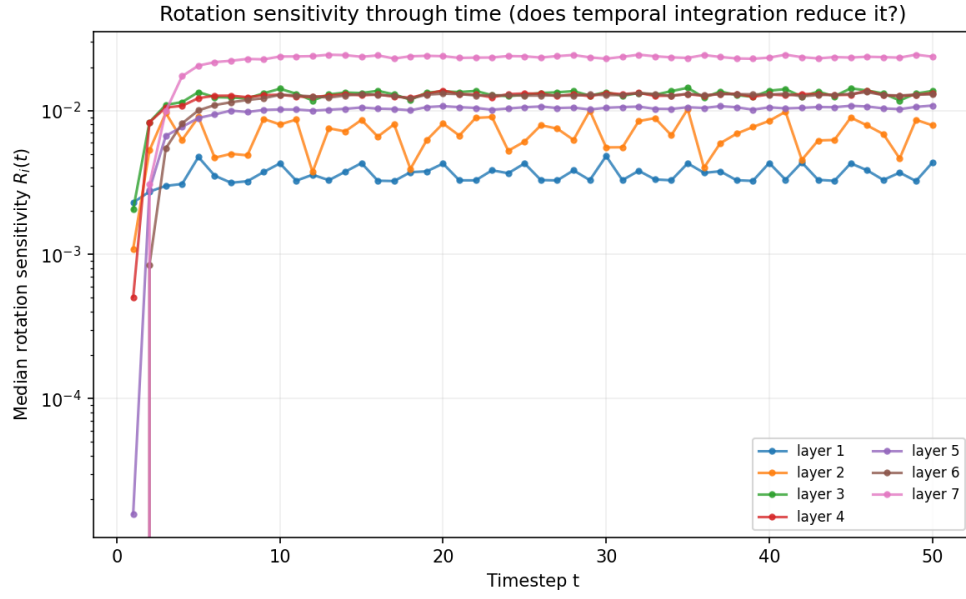


Figure 37: FULL SPIKING, seed 1. full_seed1/full_seed1_5.4_rotation_sensitivity_through_time.png

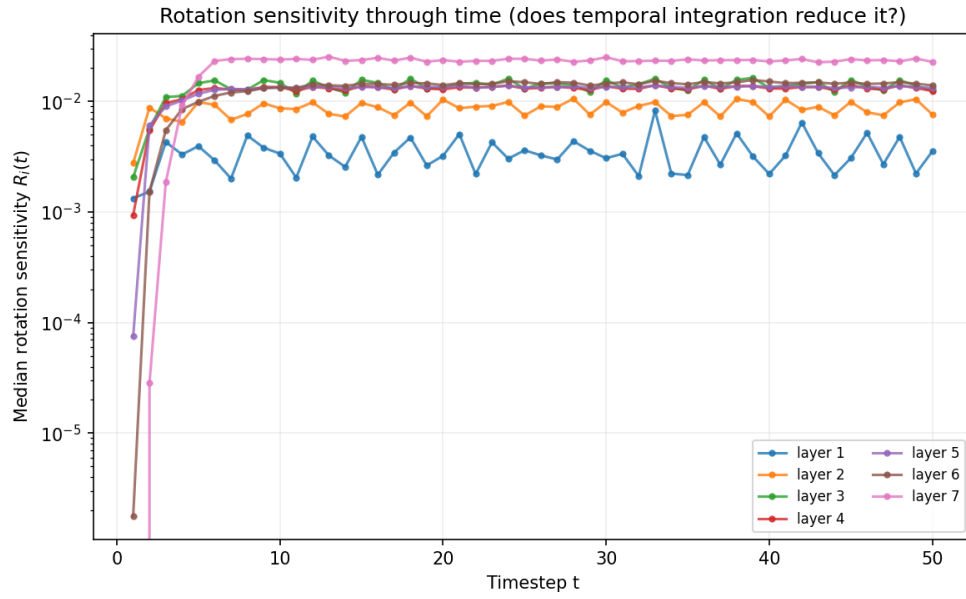


Figure 38: FULL SPIKING, seed 2. full_seed2/full_seed2_5.4_rotation_sensitivity_through_time.png

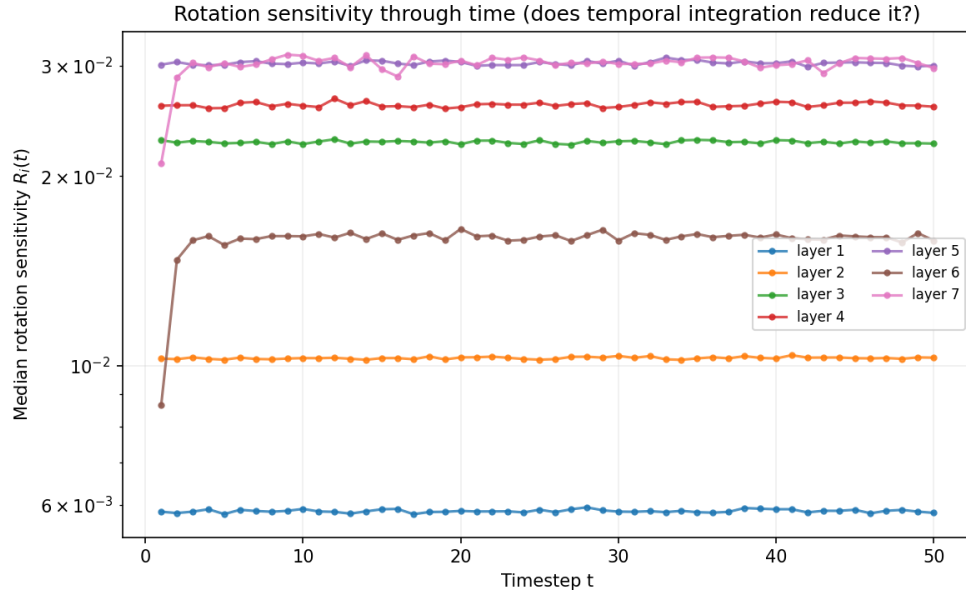


Figure 39: HYBRID LATE, seed 0. hybrid_late_seed0/hybrid_late_seed0_5.4_rotation_sensitivity_through

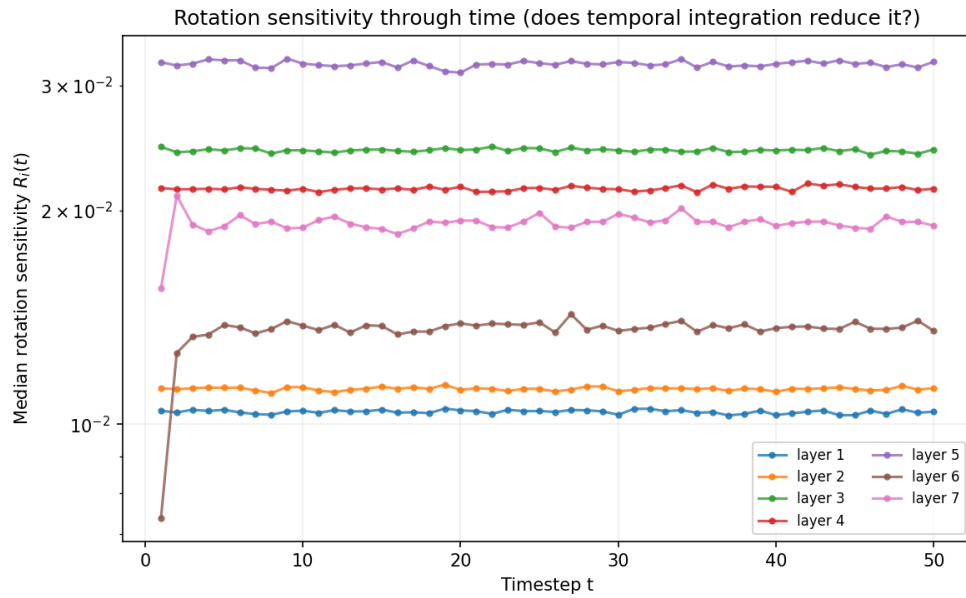


Figure 40: HYBRID LATE, seed 1. hybrid_late_seed1/hybrid_late_seed1_5.4_rotation_sensitivity_through

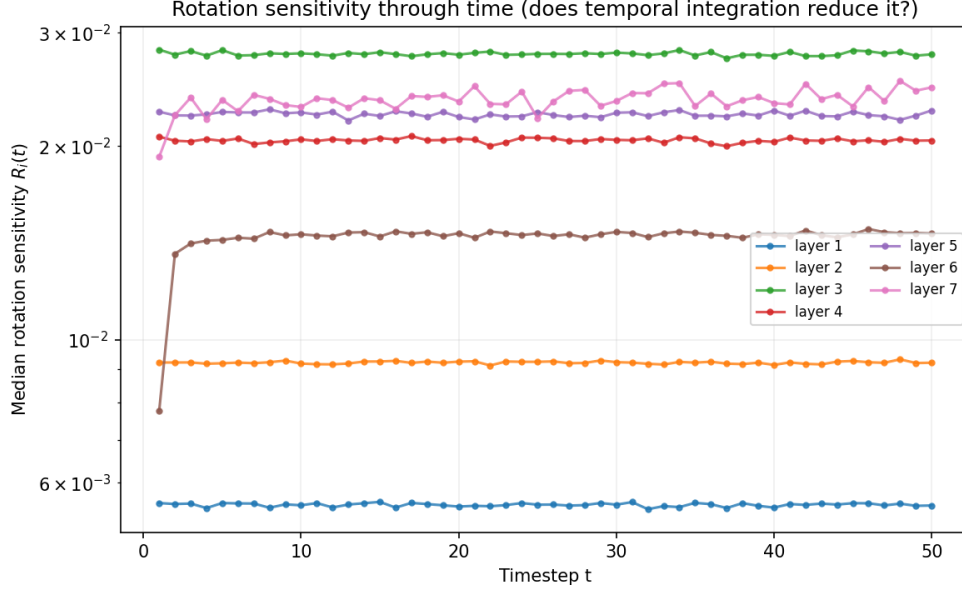


Figure 41: HYBRID LATE, seed 2. `hybrid_late_seed2/hybrid_late_seed2_5.4_rotation_sensitivity_through`

Observation 3 (Rise-then-plateau, not decay). *In every configuration and every seed, every layer’s $\mathcal{R}_i(t)$ curve rises quickly over the first 2–5 timesteps (as the membrane potential accumulates its first spikes and begins producing non-degenerate output), then plateaus for the remaining ~ 45 timesteps with only small step-to-step jitter and no visible downward trend. Layer 7 plateaus visibly higher than the other layers in ALTERNATING and FULL SPIKING (consistent with Section 5.1 of this report); in HYBRID LATE the plateau levels across layers are closer together, consistent with that configuration’s flatter layer-wise sensitivity profile.*

ANALOG has no meaningful 5.4 plot: with $T = 1$ there is no timestep axis to show evolving over time.

This directly contradicts the “temporal averaging reduces sensitivity” hypothesis of research-plan Section 9.4, as measured on these trained checkpoints: each layer’s rotation sensitivity is set almost immediately, within the first handful of timesteps, and then stays fixed for the remainder of the simulation. Temporal integration in this architecture does not act as a noise-averaging invariance-builder over the timescales examined here — the theoretical contraction property $d_{t+1} \leq \beta d_t$ proposed in research-plan Section 9.5 is not observed empirically for $\mathcal{R}_i(t)$ once the initial transient has passed.

7 Finding 3 — Temporal Evolution of Invariance via Truncated- T Accuracy (Experiment 5.7)

Section 6 examined representation-level sensitivity through time. This section asks the same question at the level that ultimately matters for a user of the classifier: accuracy. Using the truncated- T evaluation of Section 2.8, we decode a classification decision from the same trained model using only its first $T' \in \{1, 5, 10, 20, 50\}$ timesteps, and plot accuracy against all 24 test angles for each T' . This produces, for each T' , the familiar W-shaped rotation-accuracy curve (peaks at the group angles, troughs between them), and lets us observe how quickly that shape “switches on” as more

timesteps become available — directly answering the question posed in research-plan Section 5.7: does invariance emerge progressively as spikes accumulate?

7.1 Full Spiking: Requires Real Accumulation Time

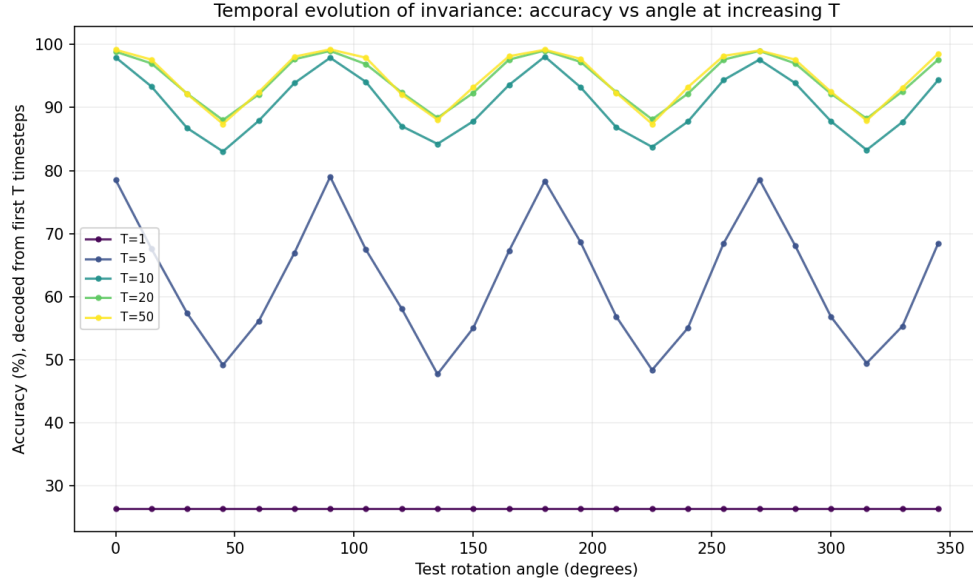


Figure 42: FULL SPIKING, seed 0 — accuracy vs. angle at $T' = 1, 5, 10, 20, 50$. `full_seed0/full_seed0_5.7_temporal_evolution.png`

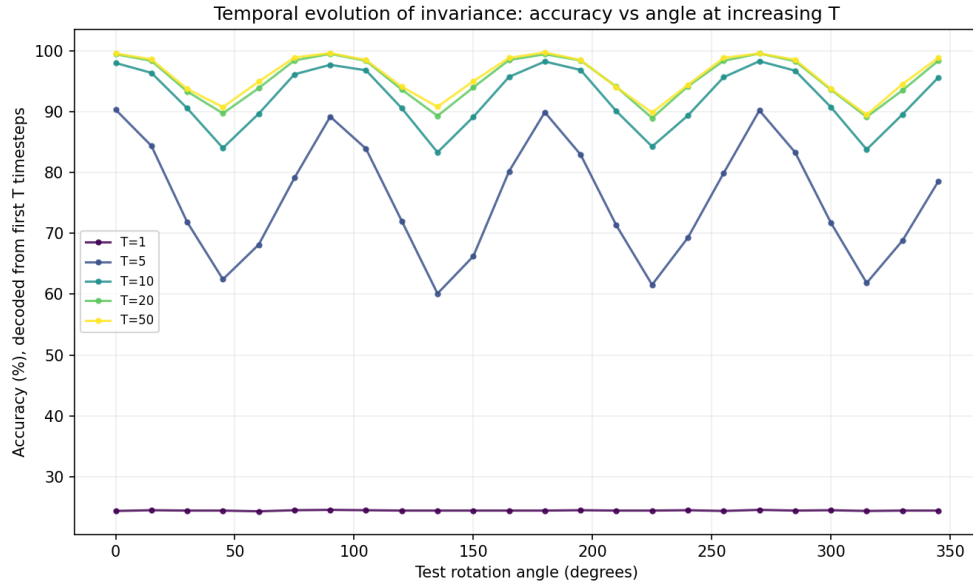


Figure 43: FULL SPIKING, seed 1. `full_seed1/full_seed1_5.7_temporal_evolution.png`

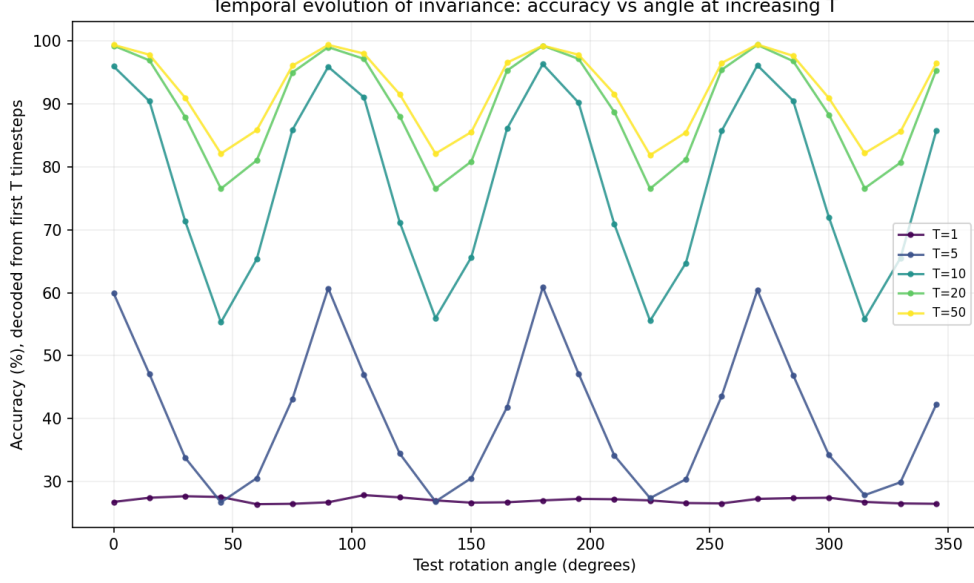


Figure 44: FULL SPIKING, seed 2. `full_seed2/full_seed2_5.7_temporal_evolution.png`

In all three seeds, the $T' = 1$ curve is flat at approximately chance level ($\sim 25\text{--}27\%$) across every angle: a single timestep carries insufficient accumulated spike information for any confident decision, correct or otherwise. Real W-shaped structure appears only by $T' = 10$, and is nearly indistinguishable from the full $T' = 50$ curve by $T' = 20$. This is mechanistically sensible: with all 7 layers spiking, every layer including the first needs several timesteps to integrate enough input spikes to pass a meaningful signal onward, and this startup delay compounds across all 7 layers.

7.2 Alternating: Partial Signal Already at $T' = 1$

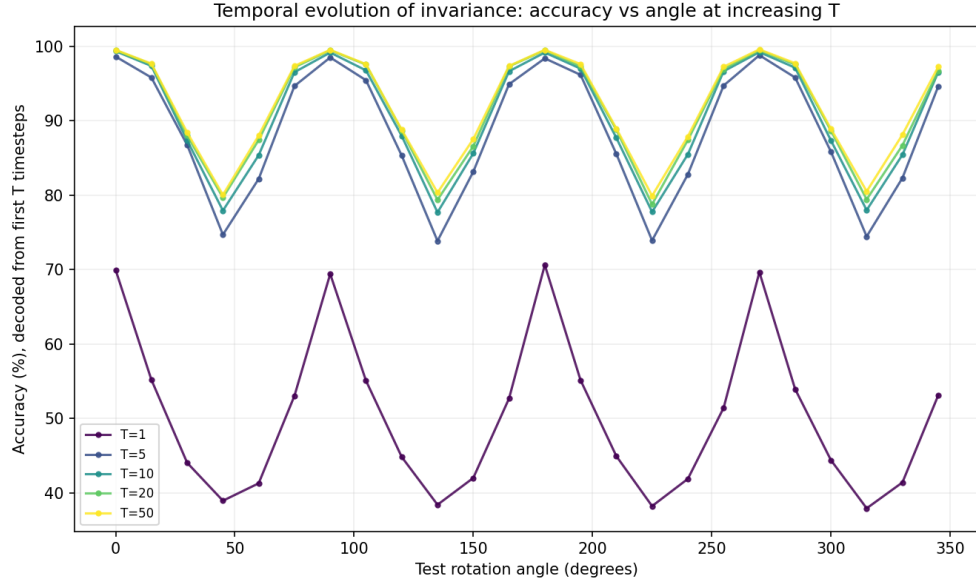


Figure 45: ALTERNATING, seed 0 — accuracy vs. angle at $T' = 1, 5, 10, 20, 50$.
alternating_seed0/alternating_seed0_5.7_temporal_evolution.png

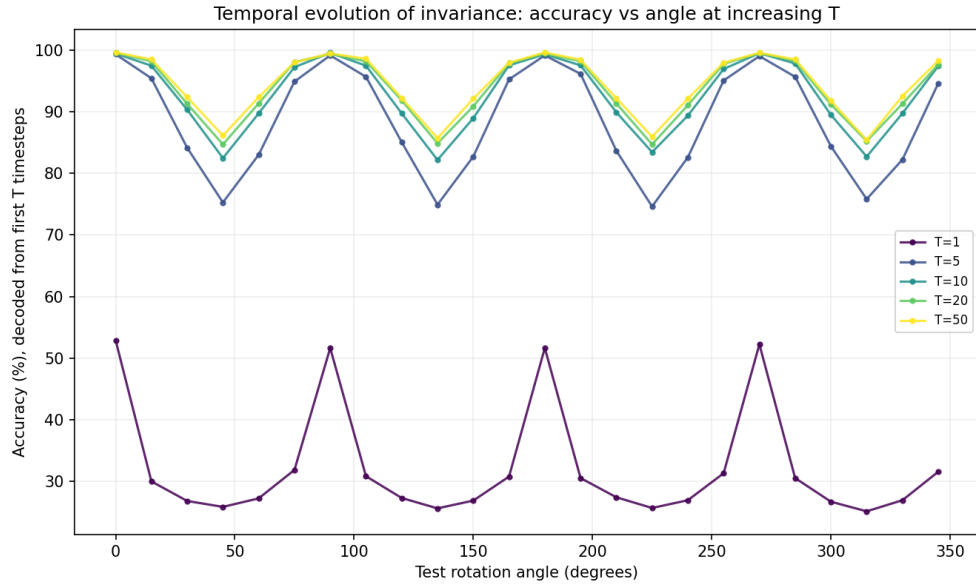


Figure 46: ALTERNATING, seed 1. alternating_seed1/alternating_seed1_5.7_temporal_evolution.png

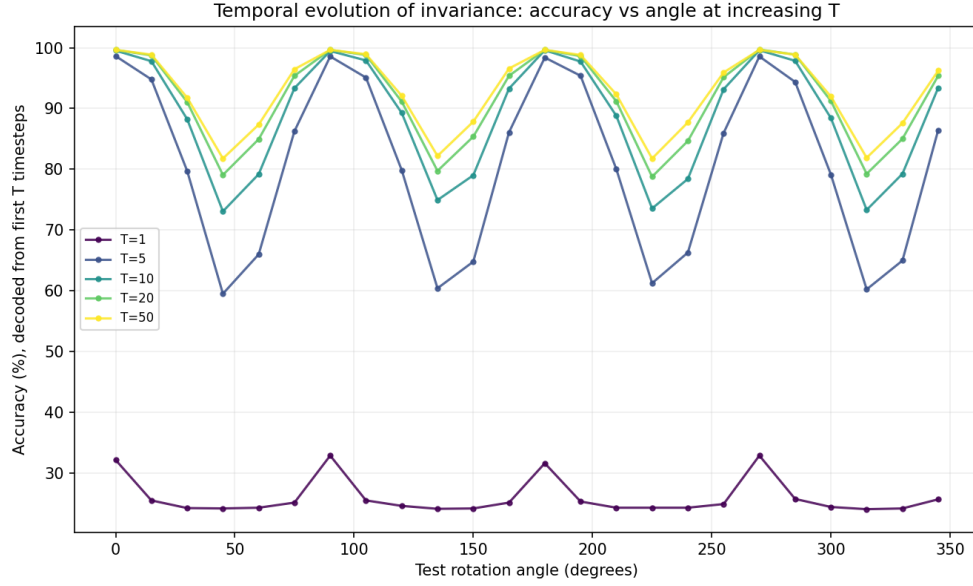


Figure 47: ALTERNATING, seed 2. `alternating_seed2/alternating_seed2_5.7_temporal_evolution.png`

Unlike FULL SPIKING, ALTERNATING’s $T' = 1$ curve is not flat: it shows a weak but genuine W-shape, oscillating roughly between 25% and 50–70% (seed- and angle-dependent), tracking the peaks and troughs of the full- T curve well below its absolute accuracy. This is architecturally sensible: with layers 1, 3, 5, 7 analog, the network has several “instant” processing stages interleaved with the spiking ones, so even one timestep of spiking input still passes through enough analog processing to carry rotation-relevant signal to the output.

7.3 Hybrid Late: Nearly Full Structure Already at $T' = 1$

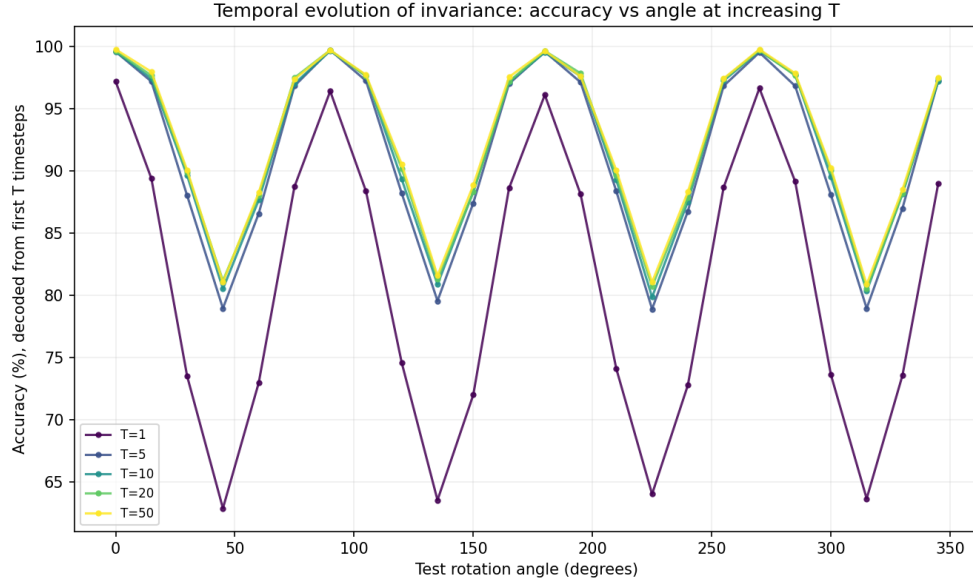


Figure 48: HYBRID LATE, seed 0 — accuracy vs. angle at $T' = 1, 5, 10, 20, 50$.
hybrid_late_seed0/hybrid_late_seed0_5.7_temporal_evolution.png

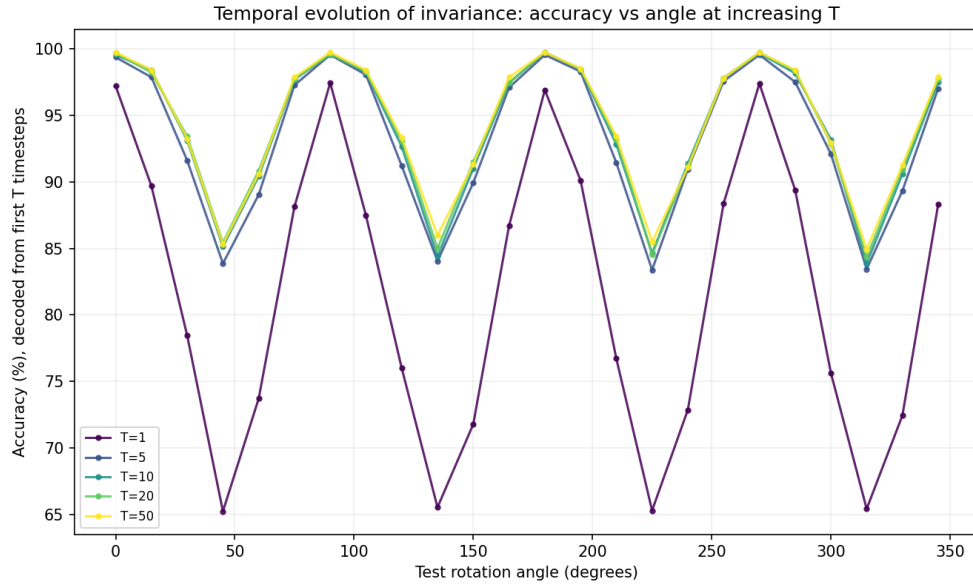


Figure 49: HYBRID LATE, seed 1. hybrid_late_seed1/hybrid_late_seed1_5.7_temporal_evolution.png

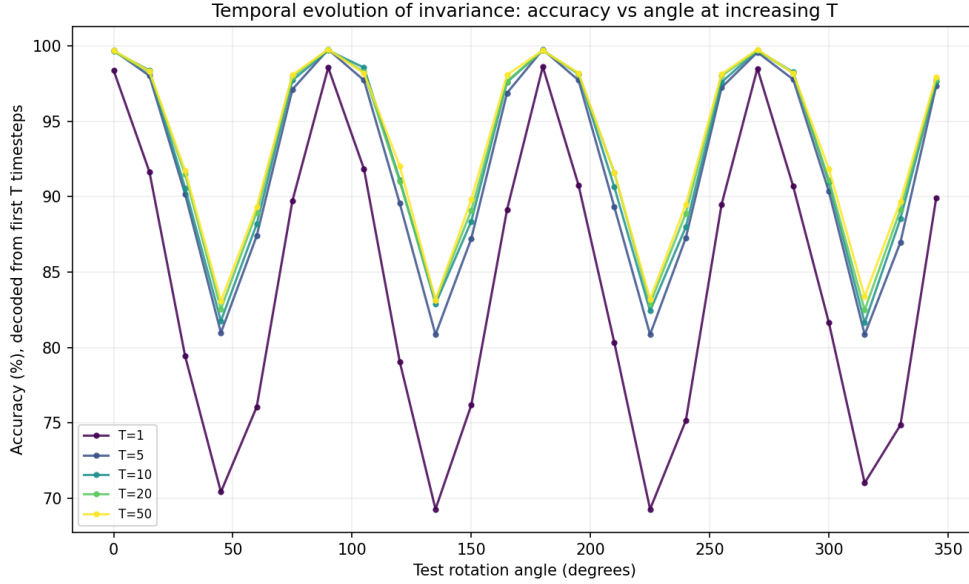


Figure 50: HYBRID LATE, seed 2. `hybrid_late_seed2/hybrid_late_seed2_5.7_temporal_evolution.png`

HYBRID LATE goes further still: its $T' = 1$ curve already closely tracks the shape of the full $T' = 50$ curve, oscillating between roughly 63% and 99% — most of the way to final performance using a single timestep. This is the expected extreme of the same pattern: with only layer 6 spiking, six of the seven layers are instantaneous analog processing, so the vast majority of the network’s rotation-relevant computation is available regardless of the number of spiking timesteps allowed.

7.4 A Clean, Reproducible Ordering, and a Compute-Budget Implication

Combining Sections 6.1–6.3 gives a consistent, monotonic ordering across all three seeds of every configuration: the more layers that spike, the more timesteps the network needs before its rotation-robust accuracy structure appears. FULL SPIKING (7/7 spiking) needs the most accumulation time, starting from true chance at $T' = 1$; ALTERNATING (3/7) needs less, starting from a partial signal; HYBRID LATE (1/7) needs almost none, starting from a strong signal already at $T' = 1$. This ordering is exactly what one would predict from counting the number of spiking “bottleneck” stages a signal must pass through before reaching the classifier, and it held identically in every one of the nine seed-runs examined (3 configurations $\times$ 3 seeds).

Observation 4 (Compute-budget implication). *For every configuration, $T' = 5$ is already close to the $T' = 50$ curve, and $T' = 10$ is nearly indistinguishable from it. Since all models here were trained and evaluated with the full $T = 50$ throughout, $T = 50$ may be substantially more compute than necessary. Future training runs — in particular the still-pending HYBRID EARLY / HYBRID MID configurations, and any controlled ablation of the original report’s $T = 30$ setting (Section 10) — could plausibly use a much lower T without materially changing outcomes, at meaningful savings in GPU wall-clock time.*

8 Finding 4 — Direct Equivariance Error (Experiment 5.8)

This section covers Experiment 5.8, the direct ground-truth measurement defined in Eq. 3 (Section 2.7): $\|f(R_\theta x) - f(x)\|/\|f(x)\|$, computed directly from the network’s final output, with no representation-similarity proxy involved. Unlike Sections 5–7, which examine internal layers, this is the most literal possible test of “does rotating the input change the answer.”

8.1 The Expected Sawtooth, Confirmed in Every Configuration

As explained in Section 2.7, the $p4$ -equivariant architecture guarantees exact (0-error) invariance only at the four group angles $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$, with no guarantee elsewhere. Every plot below confirms this precisely: equivariance error touches (or comes extremely close to) 0 at exactly the four group angles in every configuration and every seed, and rises to a local maximum roughly halfway between consecutive group angles (near $45^\circ, 135^\circ, 225^\circ, 315^\circ$), producing a repeating sawtooth of period 90° . This is exactly the behaviour the architecture’s mathematical guarantees predict, and its clean appearance in all twelve seed-runs shown here (4 configurations $\times$ 3 seeds) is a strong sanity check that the **P4BatchNorm2d** equivariance fix (Section 3.1) is functioning correctly in fully trained models, not merely in isolated unit tests.

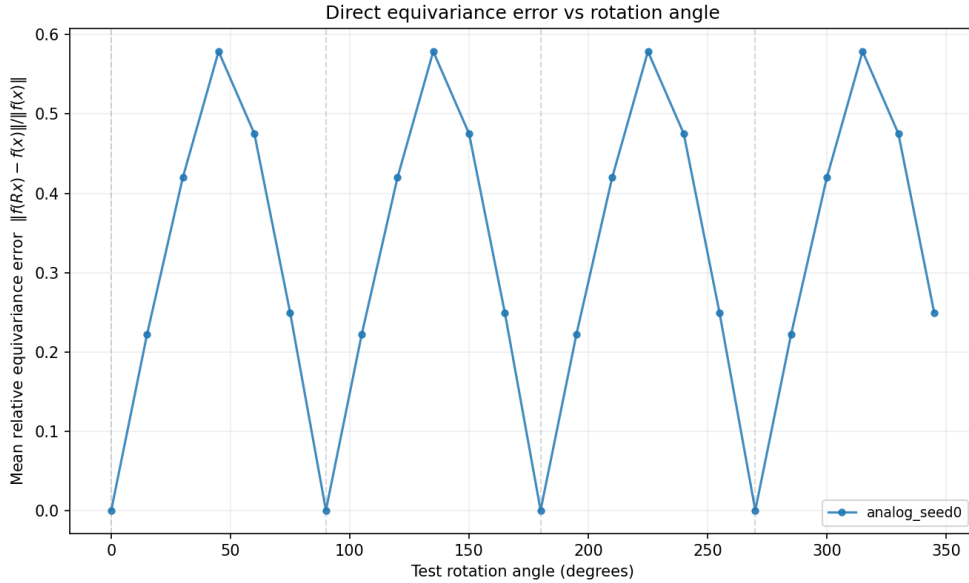


Figure 51: ANALOG, seed 0 — equivariance error vs. angle. `analog_seed0/analog_seed0_5.8_equivariance_error.png`

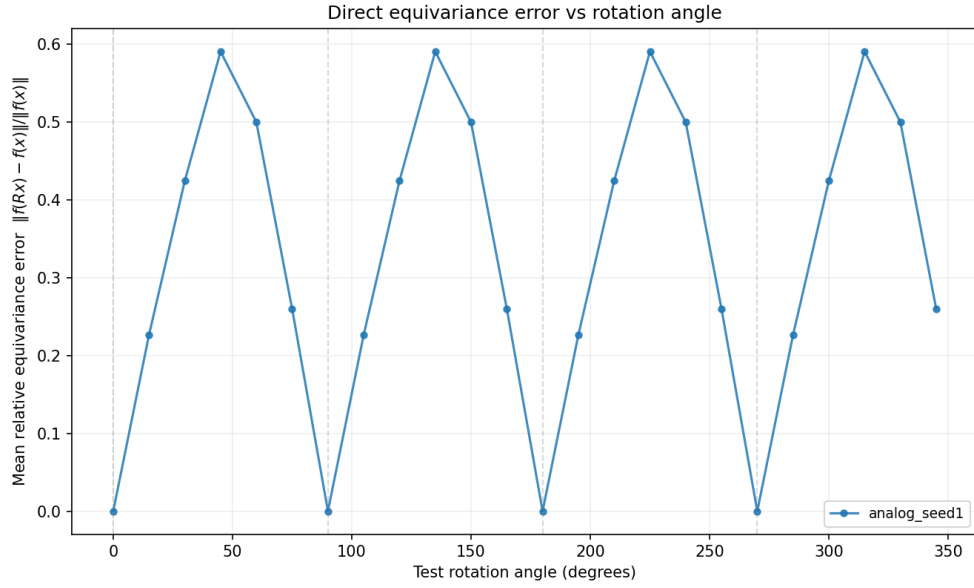


Figure 52: ANALOG, seed 1. analog_seed1/analog_seed1_5.8_equivariance_error.png

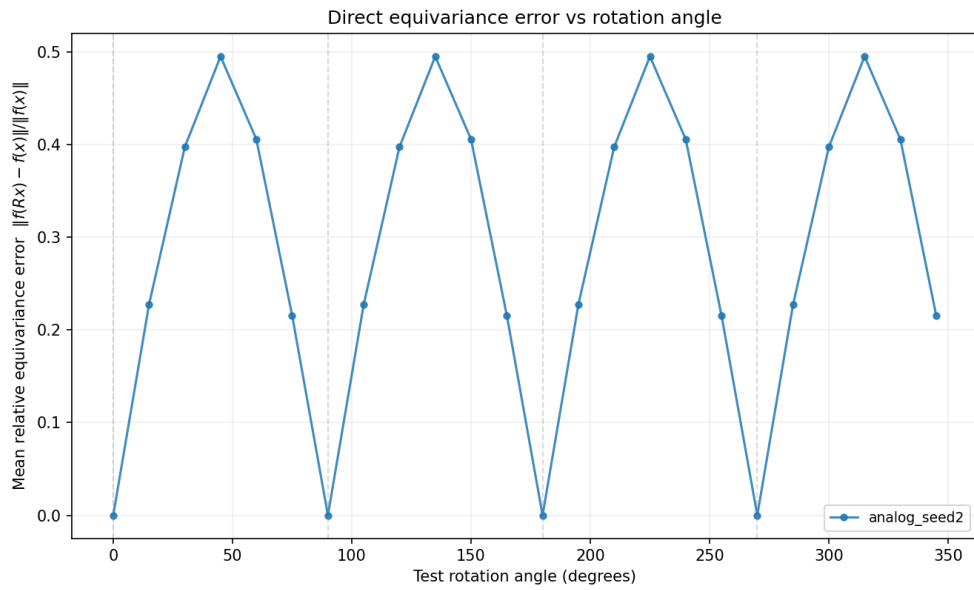


Figure 53: ANALOG, seed 2. analog_seed2/analog_seed2_5.8_equivariance_error.png

Analog



Figure 54: ALTERNATING, seed 0. alternating_seed0/alternating_seed0_5.8_equivariance_error.png



Figure 55: ALTERNATING, seed 1. alternating_seed1/alternating_seed1_5.8_equivariance_error.png

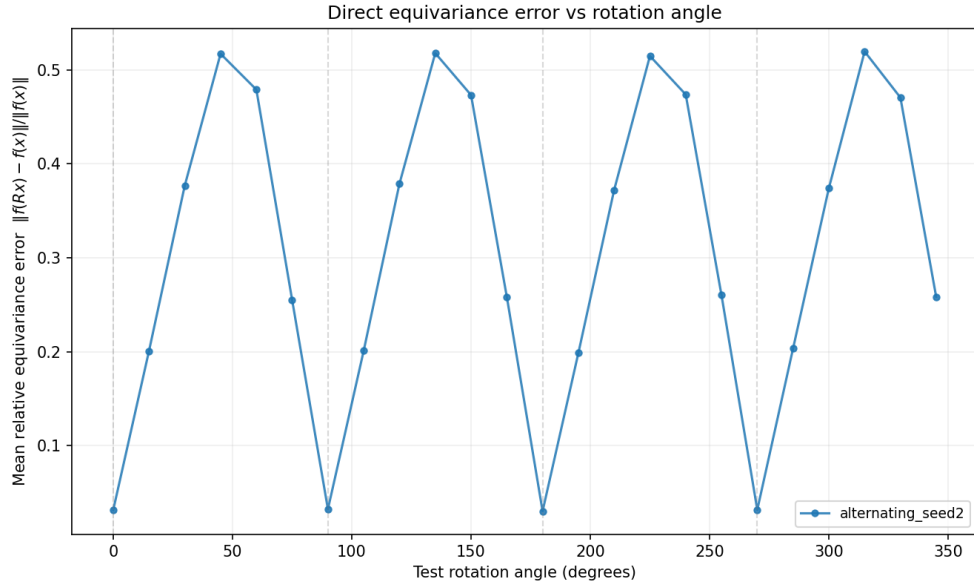


Figure 56: ALTERNATING, seed 2. `alternating_seed2/alternating_seed2_5.8_equivariance_error.png`

Alternating

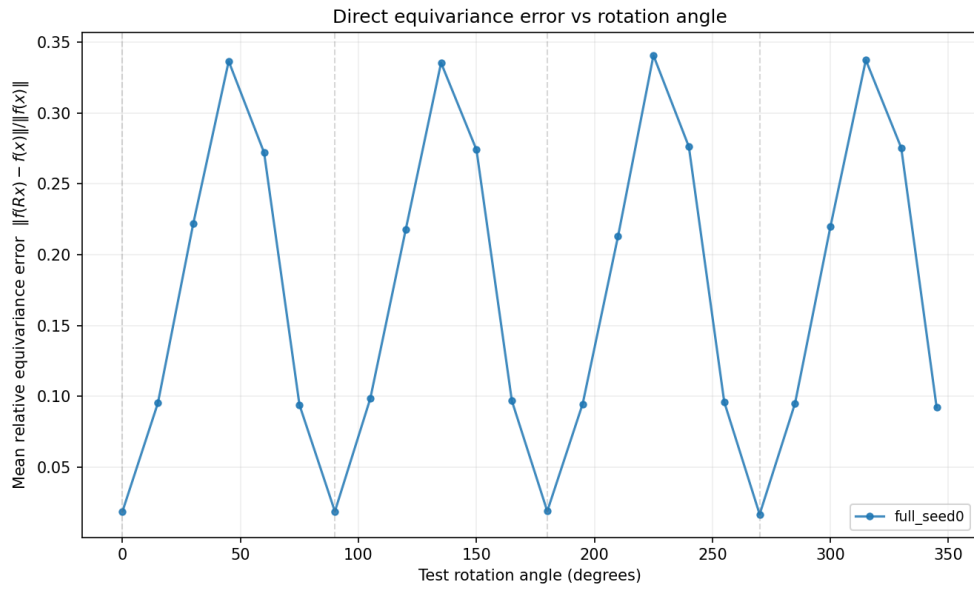


Figure 57: FULL SPIKING, seed 0. `full_seed0/full_seed0_5.8_equivariance_error.png`

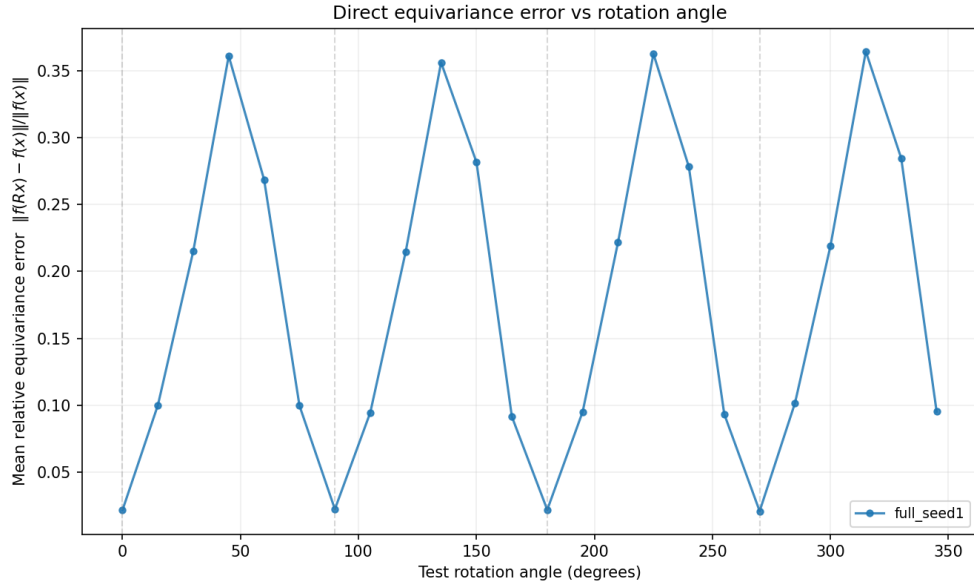


Figure 58: FULL SPIKING, seed 1. full_seed1/full_seed1_5.8_equivariance_error.png

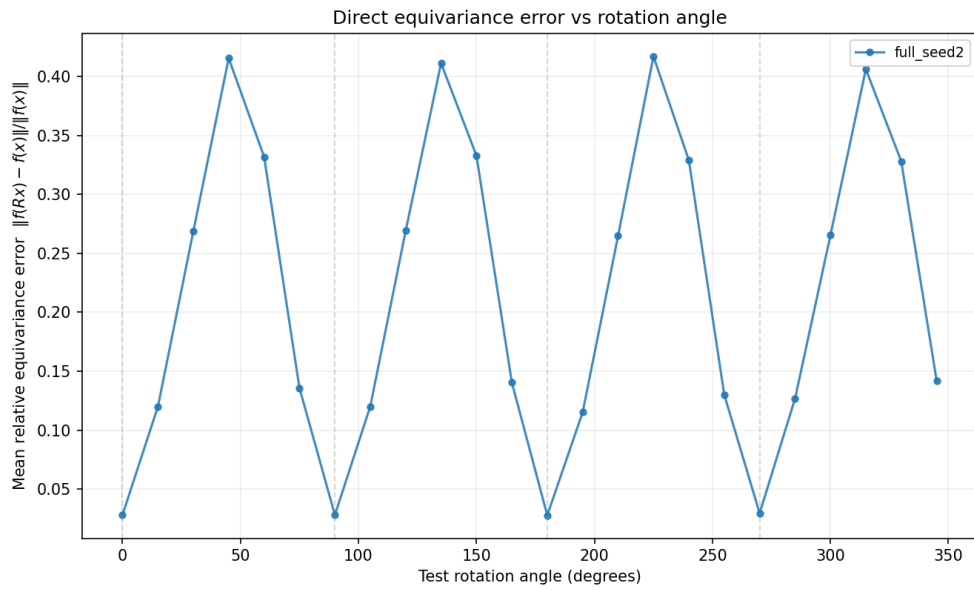


Figure 59: FULL SPIKING, seed 2. full_seed2/full_seed2_5.8_equivariance_error.png

Full Spiking

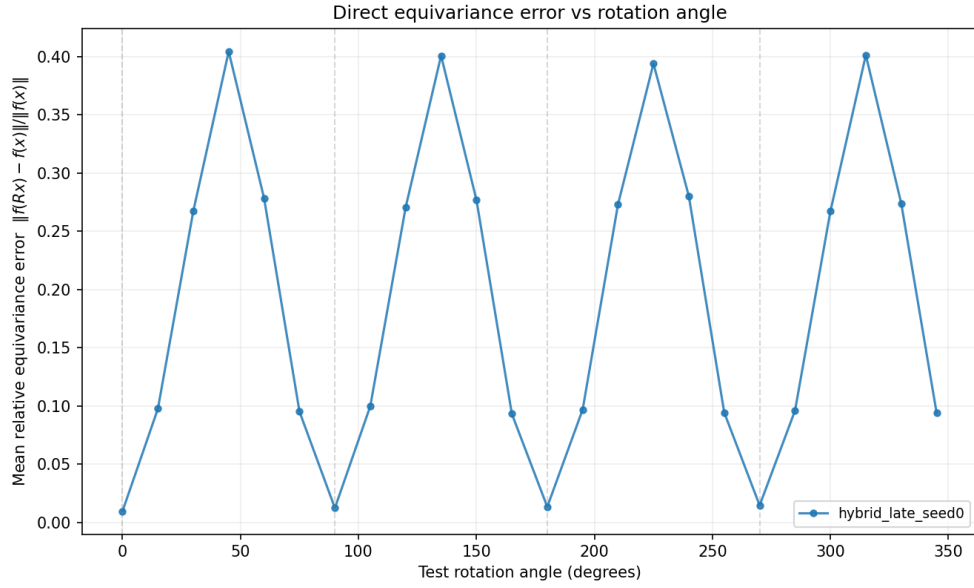


Figure 60: HYBRID LATE, seed 0. `hybrid_late_seed0/hybrid_late_seed0_5.8_equivariance_error.png`

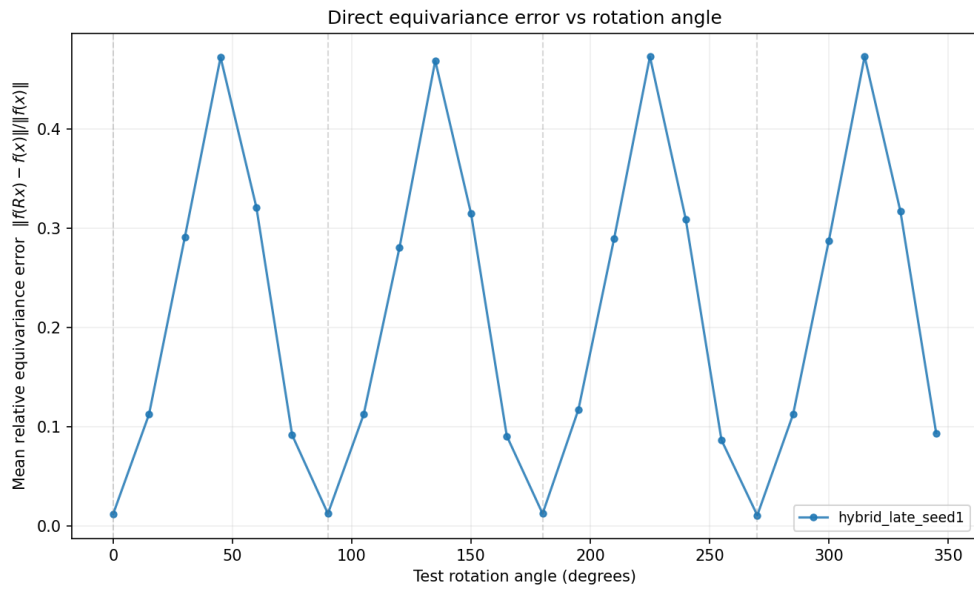


Figure 61: HYBRID LATE, seed 1. `hybrid_late_seed1/hybrid_late_seed1_5.8_equivariance_error.png`

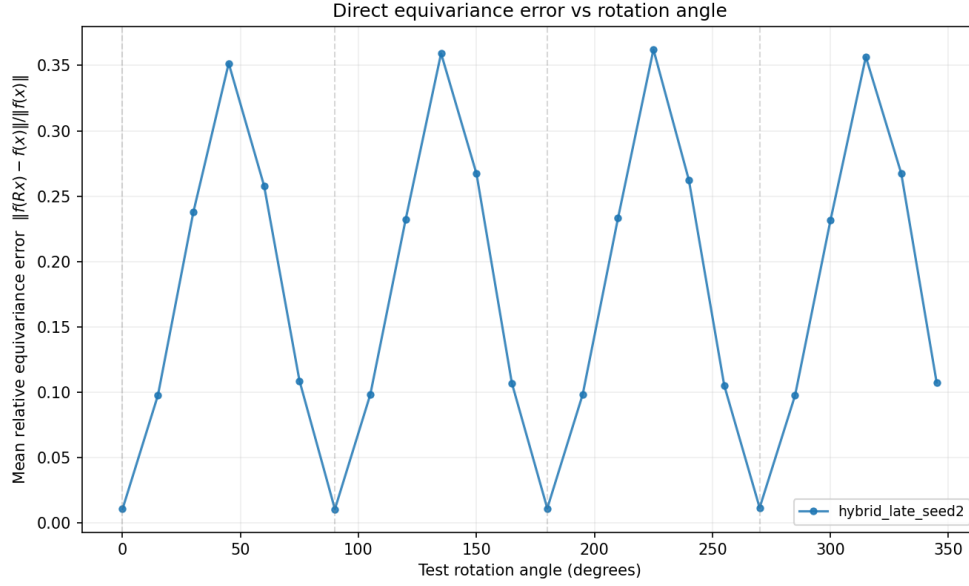


Figure 62: HYBRID LATE, seed 2. `hybrid_late_seed2/hybrid_late_seed2_5.8_equivariance_error.png`

Hybrid Late

8.2 Peak Magnitude Does Not Scale Monotonically with Spiking Extent

If spiking dynamics simply degraded invariance in proportion to how many layers spike, the sawtooth peak height (worst-case equivariance error, near 45° , 135° , 225° , 315°) should increase monotonically from ANALOG (0 spiking layers) through HYBRID LATE (1) through ALTERNATING (3) through FULL SPIKING (7). The data does not show this.

ANALOG — the zero-spiking control — has the *highest* peak error of all four configurations, and FULL SPIKING — the maximal-spiking configuration — has the *lowest*. This is the opposite of the naive “spiking hurts invariance” expectation, and is reported here as an open, reproducible observation rather than an explained result. A plausible but unconfirmed reading is that the stochastic, rate-coded nature of spiking input, combined with surrogate-gradient training, may act as an implicit regularizer that reduces the network’s worst-case output sensitivity at the hardest angles, even though (per Finding 1) internal representations are not becoming more similar by any of the similarity metrics measured. No experiment run so far isolates *why* this ordering holds.

Configuration	Spiking layers	Approx. peak equivariance error (mean over seeds)
ANALOG	0/7	$\sim 0.58\text{--}0.59$ (highest)
ALTERNATING	3/7	$\sim 0.50\text{--}0.60$
HYBRID LATE	1/7	$\sim 0.35\text{--}0.40$
FULL SPIKING	7/7	$\sim 0.33\text{--}0.42$ (lowest)

Table 3: Approximate peak equivariance error by configuration, averaged over seeds.

8.3 Interpretation Together with Finding 1

Findings 1 and 4 combine into a genuinely interesting, mildly paradoxical picture. Finding 1 showed internal representation similarity to the upright case degrades similarly across all four configurations, bottoming out at layers 6–7 regardless of spiking extent. Finding 4 shows that the output-level equivariance error not only differs meaningfully between configurations, but differs in the opposite direction from a naive “more spiking $\Rightarrow$ worse invariance” prediction. Together this means that whatever occurs in the last two layers causing representational drift does not translate linearly into output error, and spike placement affects the classifier’s output-level behaviour in a way that is not visible from intermediate-layer similarity metrics alone. This is a concrete argument, for the eventual paper, that both a representation-level analysis (Findings 1–3) and a direct output-level analysis (Finding 4) are necessary — research-plan Sections 4 and 5.8 together, neither alone tells the full story.

9 Synthesis

The four findings above were run somewhat independently but tell one coherent story when read together.

9.1 The Story in Five Statements

1. **Accuracy alone says almost nothing.** Once training is done properly (P4BatchNorm2d fix, 20 epochs, $T = 50$, 3 seeds; Section 3), all four configurations reach nearly identical rotation-accuracy curves regardless of how many layers spike, and the dramatic differences reported earlier were artifacts of the uncorrected setup, not real properties of spike placement.
2. **But representation-level analysis (Finding 1) shows the network is not internally doing the same thing in every configuration.** There is a real, consistent structural pattern — invariance holding through layers 1–5, breaking down at 6–7 — present in all configurations, including the non-spiking one. This makes it an architectural signature of the *p4*-CNN design, not a spiking signature.
3. **Individual-neuron analysis (Finding 2) shows spike placement leaves a fingerprint in the *shape* of how sensitivity builds up across depth** — ALTERNATING shows a sharp late-layer spike, FULL SPIKING and ANALOG show a smooth full-depth climb, HYBRID LATE is nearly flat — even though the endpoint (layer 7 most sensitive) is common to all. Temporal integration within a layer, however, does not smooth sensitivity out over time as research-plan Section 9.4 hypothesized; sensitivity is set within the first few timesteps and then plateaus.
4. **How quickly rotation-robust accuracy becomes available over spiking time (Finding 3) scales cleanly and reproducibly with the number of spiking layers** — more spiking layers means a longer “warm-up” before the W-shaped accuracy curve emerges. This is a clean, mechanistically explainable result the original report never surfaced, since it never varied T at fixed, correctly trained models.
5. **The direct output-level measurement (Finding 4) delivers the most surprising result:** peak equivariance error is *not* monotonic in the amount of spiking — ANALOG has the worst peak error, FULL SPIKING the best — the opposite of a naive “spiking hurts invariance” story, and unexplained pending further work.

Spike dynamics in this architecture do not determine *whether* rotation-invariant representations form — the $p4$ -equivariant architecture and its final layers do that, uniformly, regardless of spiking — but spike placement does measurably shape *how* that invariance is approached, both in depth (the per-neuron sensitivity profile across layers) and in time (how many timesteps are needed before rotation-robust accuracy appears), and, unexpectedly, does not straightforwardly worsen the network’s final output-level equivariance error as spiking is added.

9.2 Relation to the Research Plan’s Theoretical Program

Research-plan Section 9 proposes a theoretical chain,

Equivariant Spatial Layers $\implies$ Equivariant Spike Dynamics $\implies$ Equivariant Spike Counts $\implies$ Invariant Class

together with a contraction hypothesis $d_{t+1} \leq \beta d_t$ (Section 9.5 of the research plan) explaining temporal stabilization. Finding 2 (Section 6.2) provides direct empirical evidence against the contraction hypothesis as stated: $\mathcal{R}_i(t)$ does not decrease with t in any configuration tested; it plateaus. This does not invalidate the broader equivariance chain — Finding 4’s clean sawtooth pattern in every configuration is consistent with the final “Invariant Classifier” link of the chain holding at the group angles — but it does mean any future theoretical account (research-plan Sections 9.4–9.5) needs to explain a plateau, not a decay, and should treat the observed non-monotonic relationship between spiking extent and peak equivariance error (Finding 4) as a phenomenon requiring its own explanation, separate from the temporal-averaging story.

9.3 Relation to Kim et al. (2022)

Kim et al., “Exploring Temporal Information Dynamics in Spiking Neural Networks” (NeurIPS 2022), report a structurally similar decoupling in an unrelated task: their Temporal Information Concentration (TIC) finding is that Fisher Information about the input concentrates into early timesteps as training progresses, affecting robustness metrics but not final accuracy. This is the same qualitative shape as the central finding of this report — representation/robustness-level metrics differentiate configurations meaningfully while final accuracy does not — arrived at independently, in a different task and metric. This is worth citing directly in the eventual paper’s discussion as external precedent for this kind of decoupling being a real, recurring phenomenon in SNN research rather than an artifact of this project’s particular setup. It also suggests a concrete, cheap next step (Section 10): implementing Kim et al.’s Fisher Information trace I_t and Information Centroid (IC) metric on the existing checkpoints, with no retraining required, would give a single-number summary directly comparable to their own layer-wise finding (shallow layers concentrate late-timestep information, deep layers concentrate early-timestep information) against this report’s own layer-6/7 finding.

10 Limitations and Next Steps

10.1 Limitations of This Analysis

- Only 2 of the 5 spike-placement configurations from the original report’s comparison have multi-seed representation analysis alongside ANALOG, ALTERNATING, and FULL SPIKING. HYBRID EARLY and HYBRID MID have not been trained under the corrected setup, so this report says nothing about them.

- No controlled ablation has isolated which of the three candidate factors (P4BatchNorm2d fix, longer training, higher T) is responsible for the original report’s dramatic findings not replicating; all three changed simultaneously between the old and new setups.
- The counter-intuitive Finding 4 result (ANALOG worst peak equivariance error, FULL SPIKING best) has no mechanistic explanation yet; it is reported as an observed, reproducible pattern, not a settled cause-and-effect claim.
- SVCCA values should not be read as an absolute 0-to-1 similarity scale (Section 2.4) — the noise floor at the batch sizes used here is ~ 0.2 – 0.4 .
- $\mathcal{R}_i$ ’s absolute scale is not comparable across spiking vs. analog activation types (Section 6); only within-configuration comparisons across layers or seeds are on safe footing.
- All results use a single fixed task (4-class rotated MNIST, digits 3/4/8/9); generalization to the full 10-class task or other datasets is untested.

10.2 Recommended Next Steps, in Priority Order

1. Train HYBRID EARLY and HYBRID MID under the corrected setup, and extend Findings 1–4 to include them, completing the original report’s full 5-configuration picture.
2. Run the controlled ablation reproducing the original report’s 5-epoch/ $T = 30$ /CPU regime inside the corrected codebase, to isolate which factor drove the non-replication of the original report’s two dramatic findings.
3. Implement the Kim et al. Fisher Information trace I_t and Information Centroid metric on existing checkpoints (cheap, no retraining) as a precise complement to Finding 3, and cite Kim et al. directly in the paper’s discussion (Section 9).
4. Given Finding 3’s compute-budget observation ($T' = 10$ already nearly matches $T' = 50$ for every configuration tested), consider reducing T for the remaining HYBRID EARLY/HYBRID MID training runs and the ablation study above, to save GPU time on `gpuserver`.
5. Cheap, scoped additions when time allows: latent trajectory PCA plots (0° vs. 90° paths through $h_\ell(t, x)$) using the existing temporal-representation extraction infrastructure with no retraining; a narrowly scoped check of whether surrogate-gradient training keeps the equivariant weight tensor’s gradients consistent under rotation (research-plan Section 9.1).
6. Update the standalone internship report to incorporate this multi-seed representation analysis in place of the earlier seed-0-only version, and fold in the Kim et al. discussion once Step 3 is complete.

10.3 Explicitly Deprioritized (Not Recommended Before the Deadline)

- Full PLIF (learnable membrane decay, research-plan Section 5.6) retraining across all configurations — real and cheap to add to the codebase, but requires retraining every configuration again.
- DVS/N-MNIST spatio-temporal group equivariance extension — a substantially larger, separate project requiring a new dataset, new encoding, and new group-theoretic construction; not realistic before the ICLR abstract/full-paper deadlines.

- Full gated-recurrent-cell (GRU/LSTM-style) architecture extension combined with $p4$ -equivariance — genuinely novel but a multi-week undertaking requiring the same equivariance-verification rigor as the `P4BatchNorm2d` fix.
- Rotation-aware pruning (research-plan Section 5.5), learnable-threshold experiments beyond PLIF, a reduced-parameter-count architecture variant, and the 10-class task extension — all previously proposed, none run, all lower priority than completing the core 5-configuration comparison.